



Build Your Own Blocks

# 12.0

## SNAP! REFERENCE MANUAL



Brian Harvey  
Jens Mönig  
Michael Ball  
Jadga Hügler  
Victoria Phelps  
Bernat Romagosa

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## 0.1 All Snap! Blocks

| Block | Text Label                              | Type     | Category | Lisp                             |
|-------|-----------------------------------------|----------|----------|----------------------------------|
|       | move _ steps                            | command  | motion   | (move 10)                        |
|       | turn \$clockwise<br>_ degrees           | command  | motion   | (right 15)                       |
|       | turn<br>\$counterclockwise<br>_ degrees | command  | motion   | (left 15)                        |
|       | point in<br>direction _                 | command  | motion   | (head 90)                        |
|       | point towards _                         | command  | motion   | (face [mouse-<br>pointer])       |
|       | go to x: _ y: _                         | command  | motion   | (go 0 0)                         |
|       | go to _                                 | command  | motion   | (goto "[random<br>position]")    |
|       | glide _ secs to<br>x: _ y: _            | command  | motion   | (glide 1 0 0)                    |
|       | change x by _                           | command  | motion   | (+x 10)                          |
|       | set x to _                              | command  | motion   | (x= 0)                           |
|       | change y by _                           | command  | motion   | (+y 10)                          |
|       | set y to _                              | command  | motion   | (y= 0)                           |
|       | if on edge,<br>bounce                   | command  | motion   | (bounce)                         |
|       | position                                | reporter | motion   | (pos)                            |
|       | x position                              | reporter | motion   | (x)                              |
|       | y position                              | reporter | motion   | (y)                              |
|       | direction                               | reporter | motion   | (dir)                            |
|       | switch to<br>costume _                  | command  | looks    | (wear nil)                       |
|       | next costume                            | command  | looks    | (next)                           |
|       | costume #                               | reporter | looks    | (costume#)                       |
|       | say _ for _ secs                        | command  | looks    | (sayFor Hello!<br>2)             |
|       | say _                                   | command  | looks    | (say Hello!)                     |
|       | think _ for _<br>secs                   | command  | looks    | (thinkFor Hmm...<br>2)           |
|       | think _                                 | command  | looks    | (think Hmm...)                   |
|       | _ of costume _                          | reporter | looks    | (costume [width]<br>[current])   |
|       | stretch _ x: _<br>y: _ %                | reporter | looks    | (stretch<br>[current] 100<br>50) |
|       | skew _ to _<br>degrees _ %              | reporter | looks    | (skew [current]<br>0 50)         |
|       | new costume _<br>width _ height _       | reporter | looks    | (newCostume nil<br>nil nil)      |
|       | change _ effect<br>by _                 | command  | looks    | (+effect [ghost]<br>25)          |

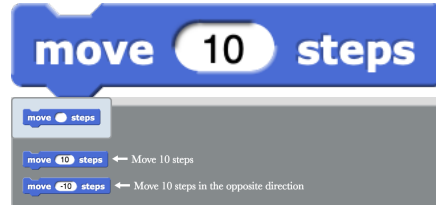
## 0.2 Motion Blocks

---

### 0.3 move steps

---

Moves a sprite a set number of steps (one coordinate) in whichever direction the sprite is facing.



#### 0.3.1 Example Images

---

Create a square shape

#### 0.3.2 Example Projects

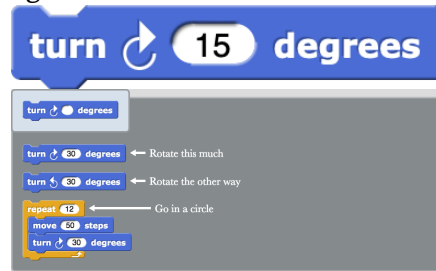
---

- Move Steps Example 1
- Move Steps Example 2
- Move Steps Example 3

## 0.4 turn ⌚ degrees

---

Turns the sprite a specified degree of angle in the clockwise direction



### 0.4.1 Example Images

---

No examples yet.

### 0.4.2 Example Projects

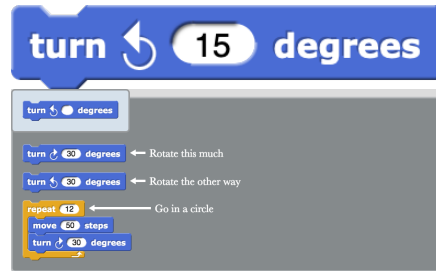
---

No examples yet.

## 0.5 turn 15 degrees

---

Complete Me



### 0.5.1 Example Images

---

No examples yet.

### 0.5.2 Example Projects

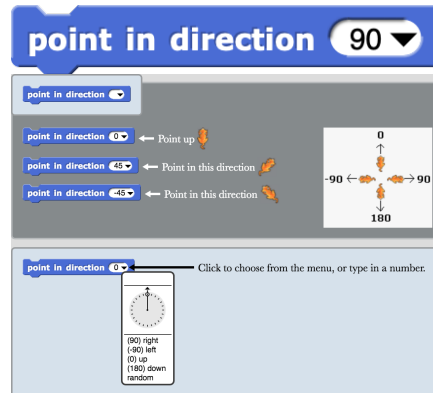
---

No examples yet.

## 0.6 point in direction

---

Complete Me



### 0.6.1 Example Images

---

No examples yet.

### 0.6.2 Example Projects

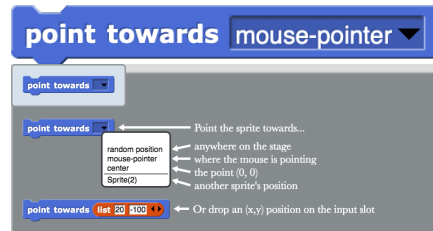
---

No examples yet.

## 0.7 point towards

---

Complete Me



### 0.7.1 Example Images

---

No examples yet.

### 0.7.2 Example Projects

---

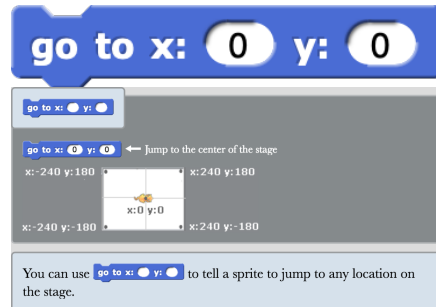
No examples yet.



## 0.8 go to x: y:

---

Complete Me



### 0.8.1 Example Images

---

No examples yet.

### 0.8.2 Example Projects

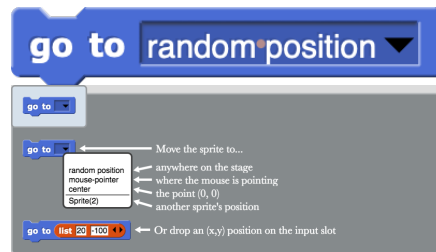
---

No examples yet.

## 0.9 go to

---

Complete Me



### 0.9.1 Example Images

---

No examples yet.

### 0.9.2 Example Projects

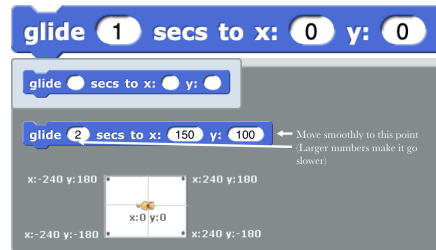
---

No examples yet.

## 0.10 Glide Secs to Position

---

Complete Me



### 0.10.1 Example Images

---

No examples yet.

### 0.10.2 Example Projects

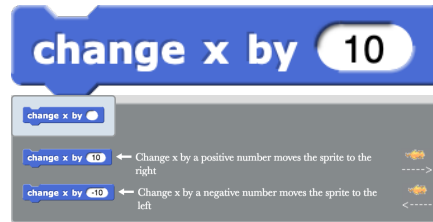
---

No examples yet.

## 0.11 change x by

---

Complete Me



### 0.11.1 Example Images

---

No examples yet.

### 0.11.2 Example Projects

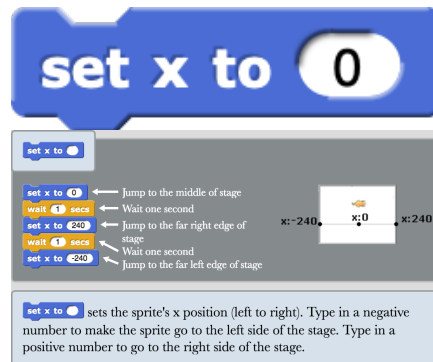
---

No examples yet.

## 0.12 set x to

---

Complete Me



### 0.12.1 Example Images

---

No examples yet.

### 0.12.2 Example Projects

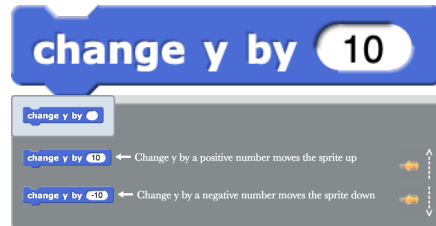
---

No examples yet.

## 0.13 change y by

---

Complete Me



### 0.13.1 Example Images

---

No examples yet.

### 0.13.2 Example Projects

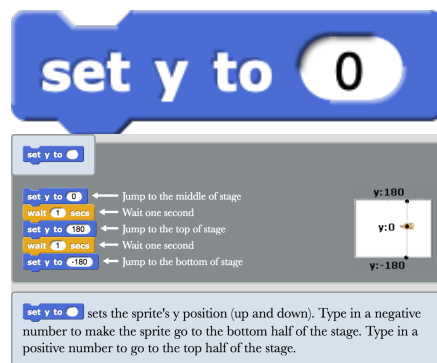
---

No examples yet.

## 0.14 set y to

---

Complete Me



### 0.14.1 Example Images

---

No examples yet.

### 0.14.2 Example Projects

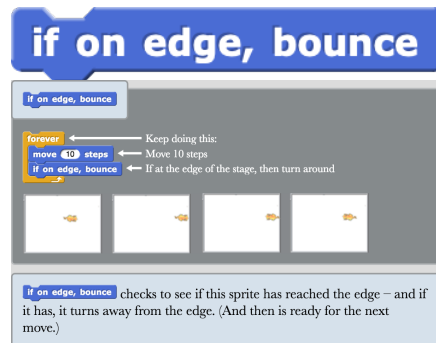
---

No examples yet.

## 0.15 if on edge, bounce

---

Complete Me



### 0.15.1 Example Images

---

No examples yet.

### 0.15.2 Example Projects

---

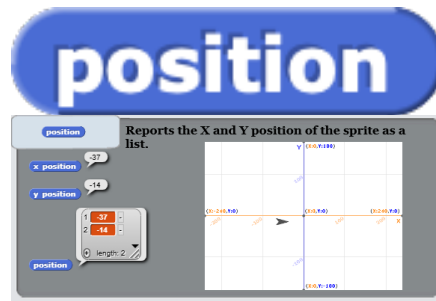
No examples yet.



## 0.16 position

---

Complete Me



### 0.16.1 Example Images

---

No examples yet.

### 0.16.2 Example Projects

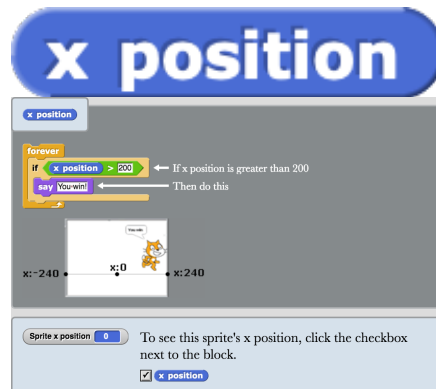
---

No examples yet.

## 0.17 x position

---

Complete Me



### 0.17.1 Example Images

---

No examples yet.

### 0.17.2 Example Projects

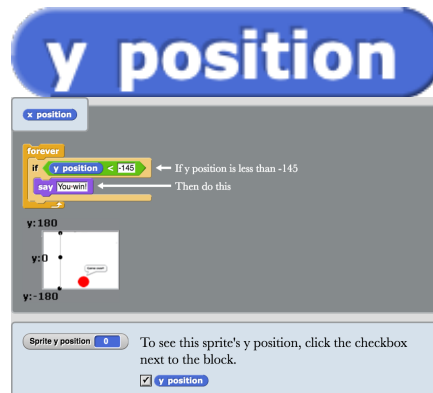
---

No examples yet.

## 0.18 y position

---

Complete Me



### 0.18.1 Example Images

---

No examples yet.

### 0.18.2 Example Projects

---

No examples yet.

## 0.19 direction

---

Complete Me



### 0.19.1 Example Images

---

No examples yet.

### 0.19.2 Example Projects

---

No examples yet.

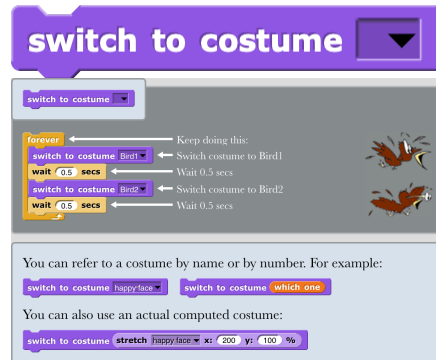
## 0.20 Looks Blocks

---

## 0.21 switch to costume

---

Complete Me



### 0.21.1 Example Images

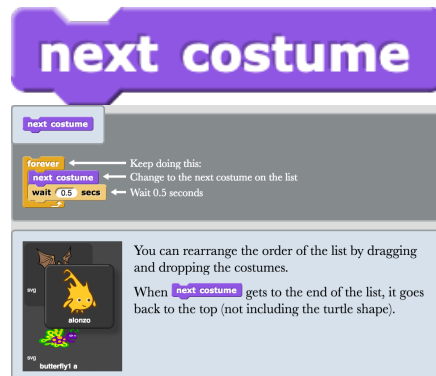
---

No examples yet.

### 0.21.2 Example Projects

---

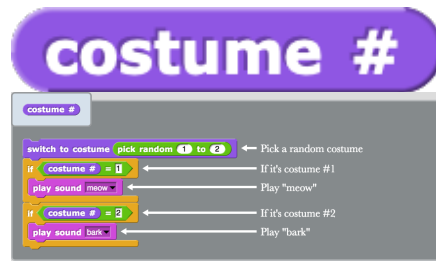
No examples yet.



## 0.23 costume number

---

Complete Me



### 0.23.1 Example Images

---

No examples yet.

### 0.23.2 Example Projects

---

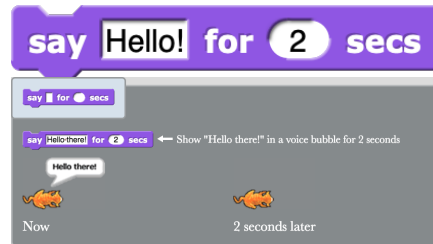
No examples yet.



## 0.24 say for secs

---

Complete Me



### 0.24.1 Example Images

---

No examples yet.

### 0.24.2 Example Projects

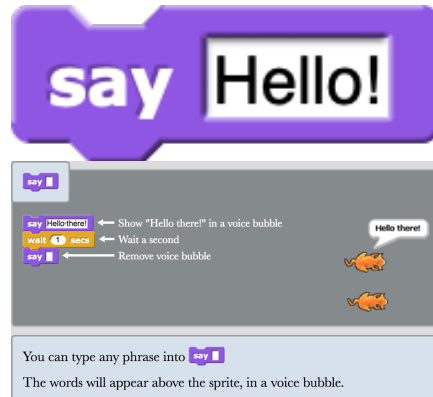
---

No examples yet.

## 0.25 say

---

Complete Me



### 0.25.1 Example Images

---

No examples yet.

### 0.25.2 Example Projects

---

No examples yet.

## 0.26 think for secs

---

Complete Me



### 0.26.1 Example Images

---

No examples yet.

### 0.26.2 Example Projects

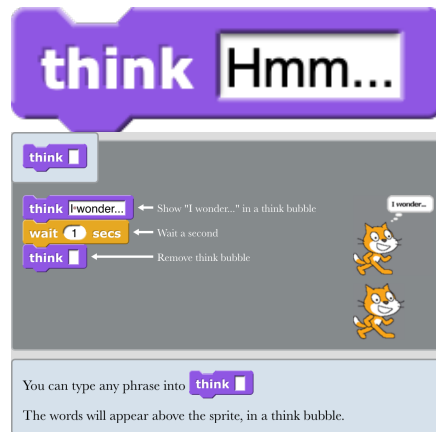
---

No examples yet.

## 0.27 think

---

Complete Me



### 0.27.1 Example Images

---

No examples yet.

### 0.27.2 Example Projects

---

No examples yet.

## 0.28 Attribute of Costume

---

Complete Me

width of costume current

width of costume

width of costume

width  
height  
pixels

Turtle  
current  
wings  
ballerina  
beard  
blue

| row | R  | G | B   | A   |
|-----|----|---|-----|-----|
| 1   | 0  | 0 | 0   | 0   |
| 2   | 0  | 0 | 0   | 0   |
| 3   | 0  | 0 | 0   | 0   |
| 4   | 0  | 0 | 0   | 0   |
| 5   | 0  | 0 | 0   | 0   |
| 6   | 0  | 0 | 0   | 0   |
| 7   | 0  | 0 | 0   | 0   |
| 8   | 45 | 0 | 0   | 255 |
| 9   | 45 | 7 | 255 | 255 |
| 10  | 45 | 8 | 255 | 255 |
| 11  | 45 | 8 | 255 | 255 |
| 12  | 45 | 8 | 255 | 255 |
| 13  | 45 | 8 | 255 | 255 |
| 14  | 45 | 8 | 255 | 255 |
| 15  | 45 | 8 | 255 | 255 |
| 16  | 45 | 8 | 255 | 255 |
| 17  | 45 | 8 | 255 | 255 |
| 18  | 45 | 8 | 255 | 255 |
| 19  | 45 | 8 | 255 | 255 |
| 20  | 45 | 8 | 255 | 255 |

pixels of costume costume

Report attributes of a costume.

Width and height are reported in pixels, but don't change in presentation mode. Pixels are reported as a list of lists, in which each sublist is the RGBA values for one pixel (Red, Green, Blue, Alpha). "Alpha" is the opacity (inverse of ghost effect). All values are between 0 and 255.

The pixel list does not have width or height information; it needs a width to make it a viewable costume. [show picture](#) in the Pixels library uses the current costume's width.

### 0.28.1 Example Images

---

No examples yet.

### 0.28.2 Example Projects

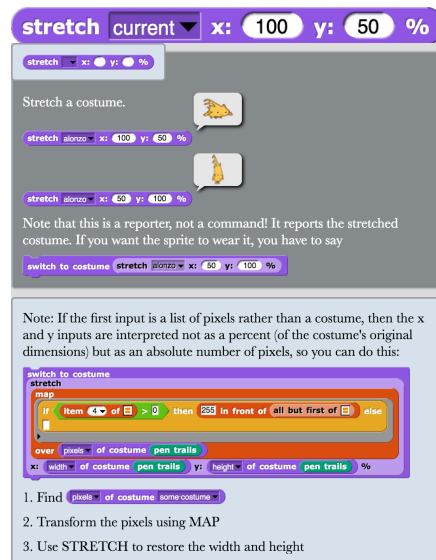
---

No examples yet.

## 0.29 Stretch Costume

---

Complete Me



The image shows a Scratch 'stretch' block with 'current' selected, x: 100, y: 50, and a percentage of 50%. Below it is a script area with two 'stretch' blocks: one with x: 100, y: 50, and another with x: 50, y: 100. A note explains that the block is a reporter, not a command, and shows a 'switch to costume' block using the 'stretch' reporter. A detailed note explains how to stretch a list of pixels using a 'map' block and 'pen trails' to restore width and height.

stretch current x: 100 y: 50 %

stretch x: 100 y: 50 %

stretch x: 50 y: 100 %

Note that this is a reporter, not a command! It reports the stretched costume. If you want the sprite to wear it, you have to say

switch to costume stretch x: 50 y: 100 %

Note: If the first input is a list of pixels rather than a costume, then the x and y inputs are interpreted not as a percent (of the costume's original dimensions) but as an absolute number of pixels, so you can do this:

switch to costume stretch

map

if item 1 of > > then 255 in front of all but first of > else 1

over pixels of costume pen trails

x: width of costume pen trails y: height of costume pen trails %

1. Find pixels of costume some costume
2. Transform the pixels using MAP
3. Use STRETCH to restore the width and height

### 0.29.1 Example Images

---

No examples yet.

### 0.29.2 Example Projects

---

No examples yet.

## 0.30 Skew Costume by Degrees

---

Complete Me

skew current▼ to 0▼ degrees 50 %

### 0.30.1 Example Images

---

No examples yet.

### 0.30.2 Example Projects

---

No examples yet.

## 0.31 New Costume

---

Complete Me



### 0.31.1 Example Images

---

No examples yet.

### 0.31.2 Example Projects

---

No examples yet.



## 0.32 change effect by

---

Complete Me



### 0.32.1 Example Images

---

No examples yet.

### 0.32.2 Example Projects

---

No examples yet.

## 0.33 set effect to

---

Complete Me



### 0.33.1 Example Images

---

No examples yet.

### 0.33.2 Example Projects

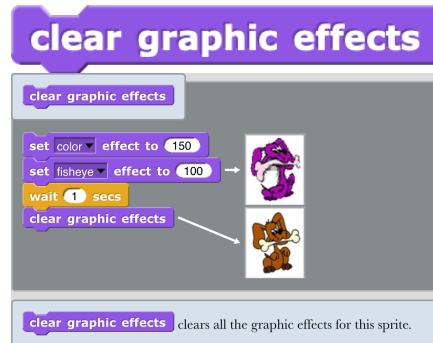
---

No examples yet.

## 0.34 clear graphic effects

---

Complete Me



### 0.34.1 Example Images

---

No examples yet.

### 0.34.2 Example Projects

---

No examples yet.

## 0.35 \_effect

---

Complete Me



### 0.35.1 Example Images

---

No examples yet.

### 0.35.2 Example Projects

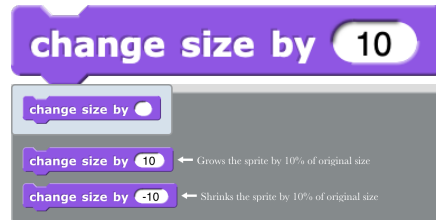
---

No examples yet.

## 0.36 change size by

---

Complete Me



### 0.36.1 Example Images

---

No examples yet.

### 0.36.2 Example Projects

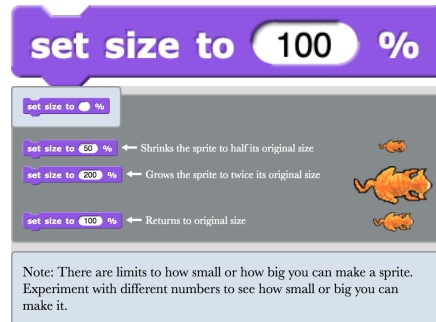
---

No examples yet.

## 0.37 Set Sprite Size

---

Complete Me



### 0.37.1 Example Images

---

No examples yet.

### 0.37.2 Example Projects

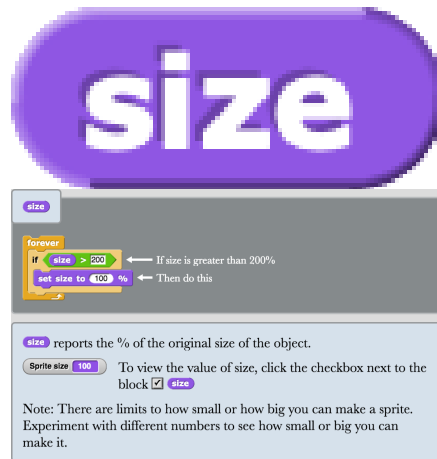
---

No examples yet.

## 0.38 size

---

Complete Me



### 0.38.1 Example Images

---

No examples yet.

### 0.38.2 Example Projects

---

No examples yet.

## 0.39 show

---

Complete Me



### 0.39.1 Example Images

---

No examples yet.

### 0.39.2 Example Projects

---

No examples yet.



## 0.40 hide

---

Complete Me



### 0.40.1 Example Images

---

No examples yet.

### 0.40.2 Example Projects

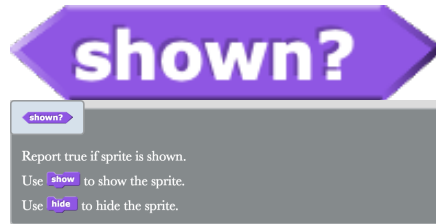
---

No examples yet.

## 0.41 shown?

---

Complete Me



### 0.41.1 Example Images

---

No examples yet.

### 0.41.2 Example Projects

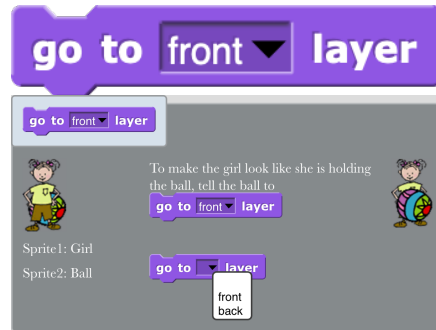
---

No examples yet.

## 0.42 go to layer

---

Complete Me



### 0.42.1 Example Images

---

No examples yet.

### 0.42.2 Example Projects

---

No examples yet.

## 0.43 go back layers

---

Complete Me



### 0.43.1 Example Images

---

No examples yet.

### 0.43.2 Example Projects

---

No examples yet.

## 0.44 Sound Blocks

---

## 0.45 Play Sound

---

Complete Me



### 0.45.1 Example Images

---

No examples yet.

### 0.45.2 Example Projects

---

No examples yet.

## 0.46 Play Sound Until Done

---

Complete Me



### 0.46.1 Example Images

---

No examples yet.

### 0.46.2 Example Projects

---

No examples yet.

## 0.47 Stop All Sounds

---

Complete Me



### 0.47.1 Example Images

---

No examples yet.

### 0.47.2 Example Projects

---

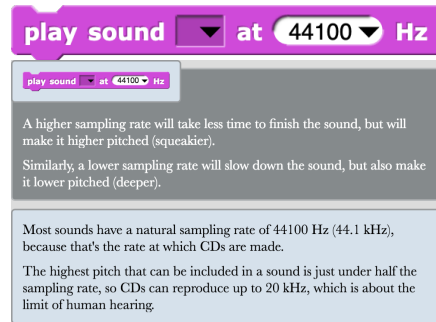
No examples yet.



## 0.48 Play Sound Hz

---

Complete Me



### 0.48.1 Example Images

---

No examples yet.

### 0.48.2 Example Projects

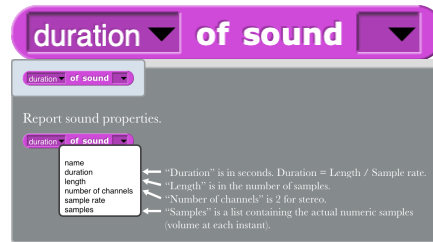
---

No examples yet.

## 0.49 Attribute Of Sound

---

Complete Me



### 0.49.1 Example Images

---

No examples yet.

### 0.49.2 Example Projects

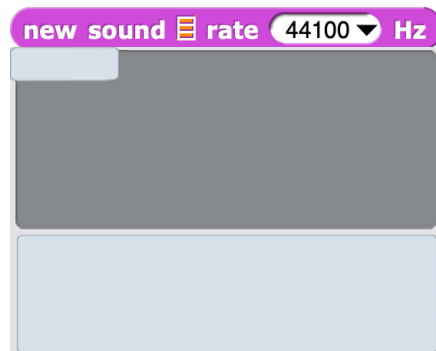
---

No examples yet.

## 0.50 New Sound Rate Hz

---

Complete Me



### 0.50.1 Example Images

---

No examples yet.

### 0.50.2 Example Projects

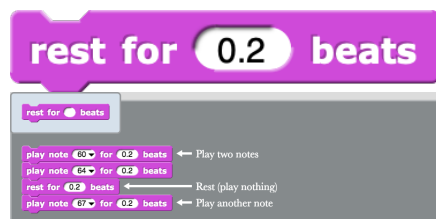
---

No examples yet.

## 0.51 Rest For Beats

---

Complete Me



### 0.51.1 Example Images

---

No examples yet.

### 0.51.2 Example Projects

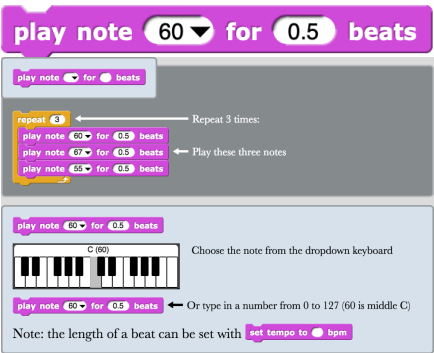
---

No examples yet.

# 0.52 Play Note For Beats

---

Complete Me



## 0.52.1 Example Images

---

No examples yet.

## 0.52.2 Example Projects

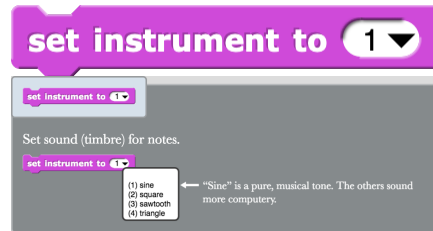
---

No examples yet.

## 0.53 Set Instrument

---

Complete Me



### 0.53.1 Example Images

---

No examples yet.

### 0.53.2 Example Projects

---

No examples yet.

# 0.54 Change Tempo

---

Complete Me

change tempo by20

change tempo by

set tempo to60bpm

Set tempo to 60 beats per minute

forever

Keep doing this

play noteC4for0.5beats

Play a note for 0.5 beats

change tempo by20

Increase the tempo by 20 beats

iftempo>500

If the tempo is greater than 500

stop this script

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play noteC4for0.5beats

0.5 beats = 0.5 seconds at 60 bpm

play noteC4for0.2beats

0.2 beats = 0.1 seconds at 120 bpm

## 0.54.1 Example Images

---

No examples yet.

## 0.54.2 Example Projects

---

No examples yet.

# 0.55 Set Tempo

---

Complete Me

set tempo to 60 bpm

set tempo to 60 bpm

forever

play note C4 for 0.5 beats

change tempo by 20

if tempo > 500

stop this script

Set tempo to 60 beats per minute

Keep doing this

Play a note for 0.5 beats

Increase the tempo by 20 beats

If the tempo is greater than 500

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play note C4 for 0.5 beats ← 0.5 beats = 0.5 seconds at 60 bpm

play note C4 for 0.2 beats ← 0.2 beats = 0.1 seconds at 120 bpm

## 0.55.1 Example Images

---

No examples yet.

## 0.55.2 Example Projects

---

No examples yet.



# 0.56    Tempo

---

Complete Me

tempo

set tempo to 60 bpm

Set tempo to 60 beats per minute

forever

Keep doing this:

play note 64 for 0.5 beats

Play a note for 0.5 beats

change tempo by 20

Increase the tempo by 20 beats

if tempo > 500

If the tempo is greater than 500

stop this script

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play note 60 for 0.5 beats

0.5 beats = 0.5 seconds at 60 bpm

play note 65 for 0.2 beats

0.2 beats = 0.1 seconds at 120 bpm

## 0.56.1    Example Images

---

No examples yet.

## 0.56.2    Example Projects

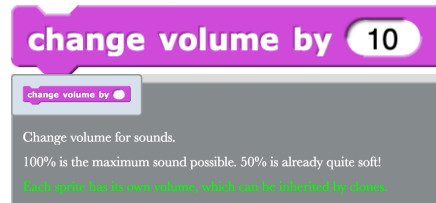
---

No examples yet.

## 0.57 Change Volume

---

Complete Me



### 0.57.1 Example Images

---

No examples yet.

### 0.57.2 Example Projects

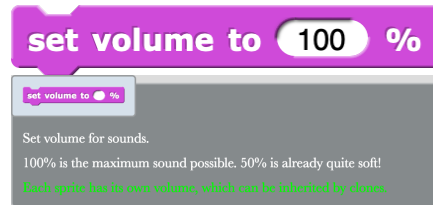
---

No examples yet.

## 0.58 Set Volume

---

Complete Me



### 0.58.1 Example Images

---

No examples yet.

### 0.58.2 Example Projects

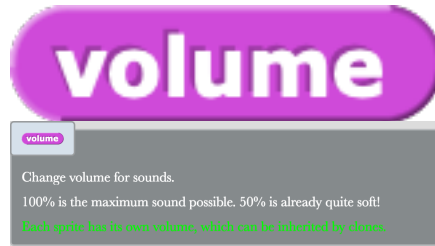
---

No examples yet.

## 0.59 Volume

---

Complete Me



### 0.59.1 Example Images

---

No examples yet.

### 0.59.2 Example Projects

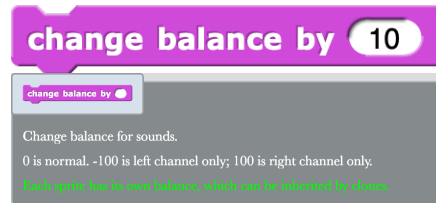
---

No examples yet.

## 0.60 Change Balance

---

Complete Me



### 0.60.1 Example Images

---

No examples yet.

### 0.60.2 Example Projects

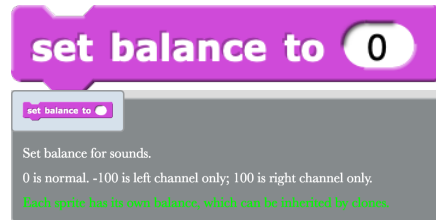
---

No examples yet.

## 0.61 Set Balance To

---

Complete Me



### 0.61.1 Example Images

---

No examples yet.

### 0.61.2 Example Projects

---

No examples yet.

## 0.62 Report Balance

---

Complete Me



### 0.62.1 Example Images

---

No examples yet.

### 0.62.2 Example Projects

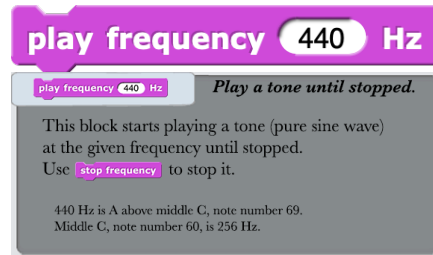
---

No examples yet.

## 0.63 Play Frequency Hz

---

Complete Me



### 0.63.1 Example Images

---

No examples yet.

### 0.63.2 Example Projects

---

No examples yet.



## 0.64 Stop Frequency

---

Complete Me



### 0.64.1 Example Images

---

No examples yet.

### 0.64.2 Example Projects

---

No examples yet.



## 0.66 clear

---

Complete Me



### 0.66.1 Example Images

---

No examples yet.

### 0.66.2 Example Projects

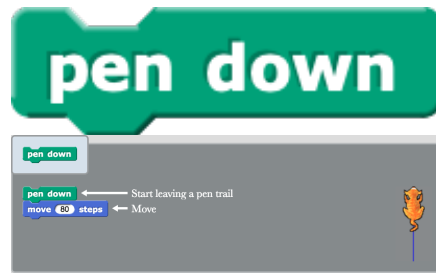
---

No examples yet.

## 0.67 pen down

---

Complete Me



### 0.67.1 Example Images

---

No examples yet.

### 0.67.2 Example Projects

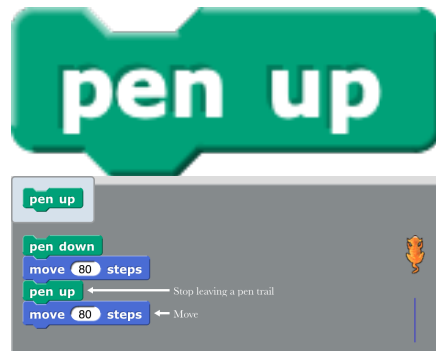
---

No examples yet.

## 0.68 pen up

---

Complete Me



### 0.68.1 Example Images

---

No examples yet.

### 0.68.2 Example Projects

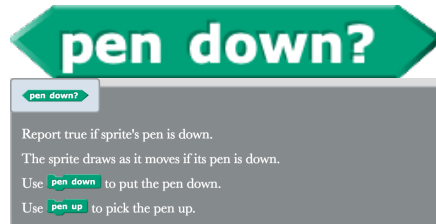
---

No examples yet.

## 0.69 pen down?

---

Complete Me



### 0.69.1 Example Images

---

No examples yet.

### 0.69.2 Example Projects

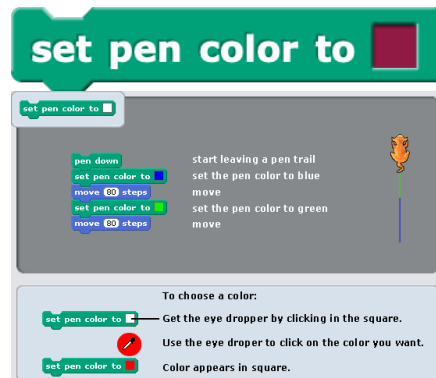
---

No examples yet.

## 0.70 set pen color to

---

Complete Me



### 0.70.1 Example Images

---

No examples yet.

### 0.70.2 Example Projects

---

No examples yet.

## 0.71 change pen by

---

Complete Me



### 0.71.1 Example Images

---

No examples yet.

### 0.71.2 Example Projects

---

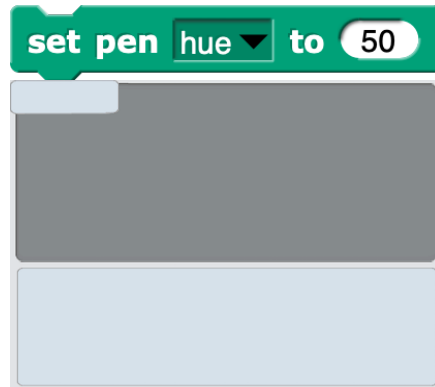
No examples yet.



## 0.72 set pen to

---

Complete Me



### 0.72.1 Example Images

---

No examples yet.

### 0.72.2 Example Projects

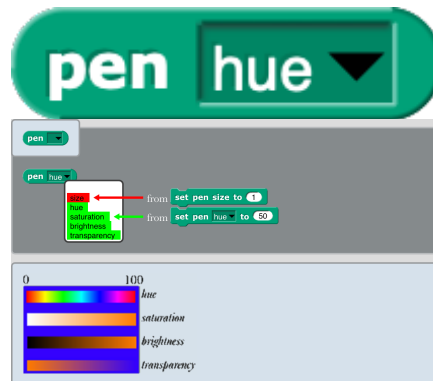
---

No examples yet.

## 0.73 pen

---

Complete Me



### 0.73.1 Example Images

---

No examples yet.

### 0.73.2 Example Projects

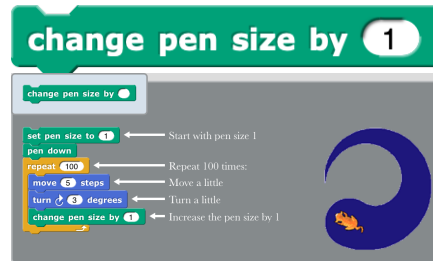
---

No examples yet.

## 0.74 change pen size by

---

Complete Me



### 0.74.1 Example Images

---

No examples yet.

### 0.74.2 Example Projects

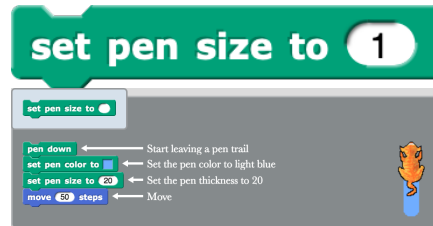
---

No examples yet.

## 0.75 set pen size to

---

Complete Me



### 0.75.1 Example Images

---

No examples yet.

### 0.75.2 Example Projects

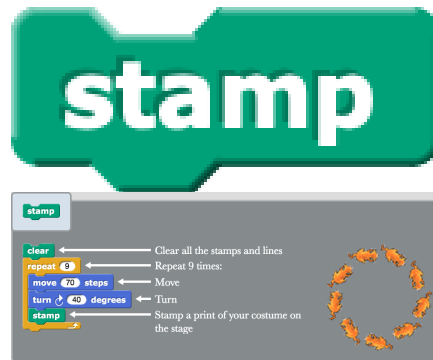
---

No examples yet.

## 0.76 stamp

---

Complete Me



### 0.76.1 Example Images

---

No examples yet.

### 0.76.2 Example Projects

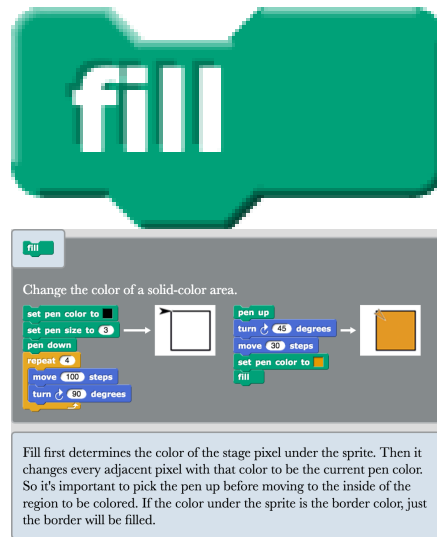
---

No examples yet.

## 0.77 fill

---

Complete Me



### 0.77.1 Example Images

---

No examples yet.

### 0.77.2 Example Projects

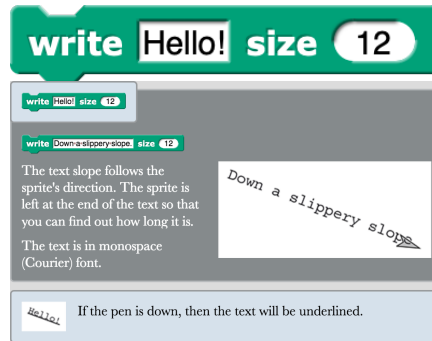
---

No examples yet.

## 0.78 write size

---

Complete Me



### 0.78.1 Example Images

---

No examples yet.

### 0.78.2 Example Projects

---

No examples yet.

0.79 pen trails

---

Complete Me

pen trails

Report lines drawn on stage as a costume.



pen trails

The reported value does not include the stage background or the sprite itself, just the lines drawn with the pen.

What can you do with the reported costume?

Add it to your costumes. 

add pen trails to my costumes

Modify it. 

switch to costume

stretch

map

if item of then then in front of all but first of else

over of costume pen trails

of costume pen trails of costume pen trails %

0.79.1 Example Images

---

No examples yet.

0.79.2 Example Projects

---

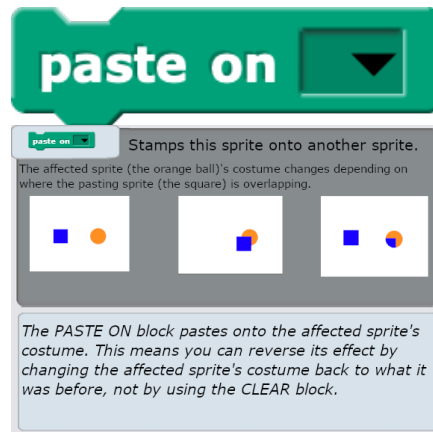
No examples yet.



## 0.80 paste on

---

Complete Me



### 0.80.1 Example Images

---

No examples yet.

### 0.80.2 Example Projects

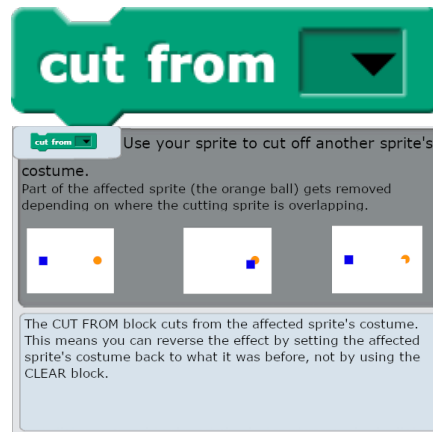
---

No examples yet.

## 0.81 cut from

---

Complete Me



### 0.81.1 Example Images

---

No examples yet.

### 0.81.2 Example Projects

---

No examples yet.

## 0.82 Color

---

Complete Me



### 0.82.1 Example Images

---

No examples yet.

### 0.82.2 Example Projects

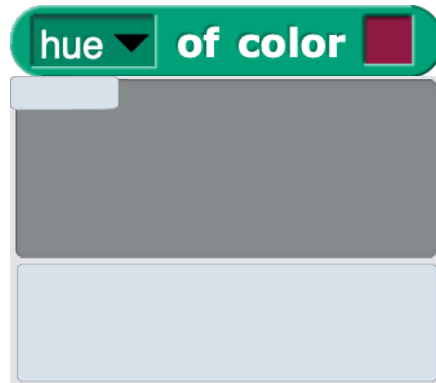
---

No examples yet.

## 0.83 Color Attribute

---

Complete Me



### 0.83.1 Example Images

---

No examples yet.

### 0.83.2 Example Projects

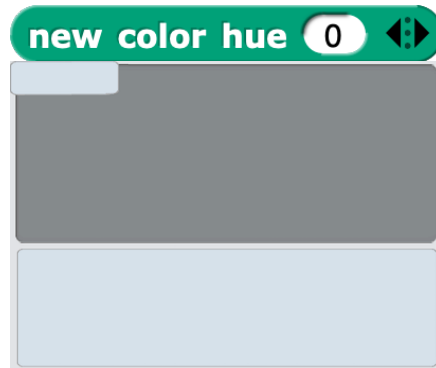
---

No examples yet.

## 0.84 new color

---

Complete Me



### 0.84.1 Example Images

---

No examples yet.

### 0.84.2 Example Projects

---

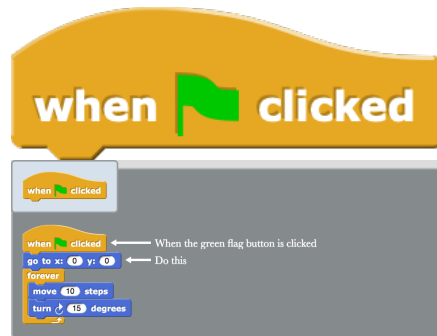
No examples yet.



## 0.86 When Green Flag Clicked

---

Complete Me



### 0.86.1 Example Images

---

No examples yet.

### 0.86.2 Example Projects

---

No examples yet.

## 0.87 when is edited

---

Complete Me



### 0.87.1 Example Images

---

No examples yet.

### 0.87.2 Example Projects

---

No examples yet.



## 0.88 when I start as a clone

---

Complete Me

### when I start as a clone

when I start as a clone

Clone initialization script

when clicked  
repeat 3  
create a clone of Sprite

→

(because the original and all three copies are in the same place and the same color)

when I start as a clone  
set pen hue to pick random 1 to 100  
set pen saturation to 100  
set pen brightness to 100  
go to x: pick random -50 to 50 y: pick random -50 to 50

when clicked  
repeat 3  
create a clone of Sprite

→

Note: Snap! clones are not copies of the sprite; they actually share information, so a change in the original may be seen in the clones. See the “Object Oriented Programming with Sprites” chapter in the Snap! Reference Manual.

### 0.88.1 Example Images

---

No examples yet.

### 0.88.2 Example Projects

---

No examples yet.

## 0.89 when I receive

---

Complete Me



### 0.89.1 Example Images

---

No examples yet.

### 0.89.2 Example Projects

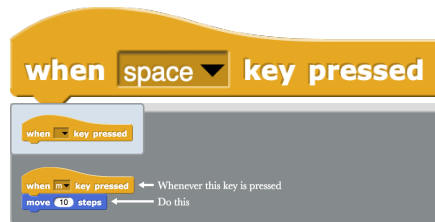
---

No examples yet.

## 0.90 when key pressed

---

Complete Me



### 0.90.1 Example Images

---

No examples yet.

### 0.90.2 Example Projects

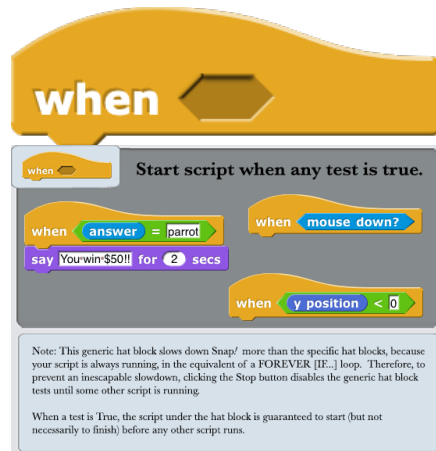
---

No examples yet.

## 0.91 when

---

Complete Me



### 0.91.1 Example Images

---

No examples yet.

### 0.91.2 Example Projects

---

No examples yet.

## 0.92 when I am

---

Complete Me



### 0.92.1 Example Images

---

No examples yet.

### 0.92.2 Example Projects

---

No examples yet.

## 0.93 broadcast

---

Complete Me



### 0.93.1 Example Images

---

No examples yet.

### 0.93.2 Example Projects

---

No examples yet.

## 0.94 broadcast and wait

---

Complete Me



### 0.94.1 Example Images

---

No examples yet.

### 0.94.2 Example Projects

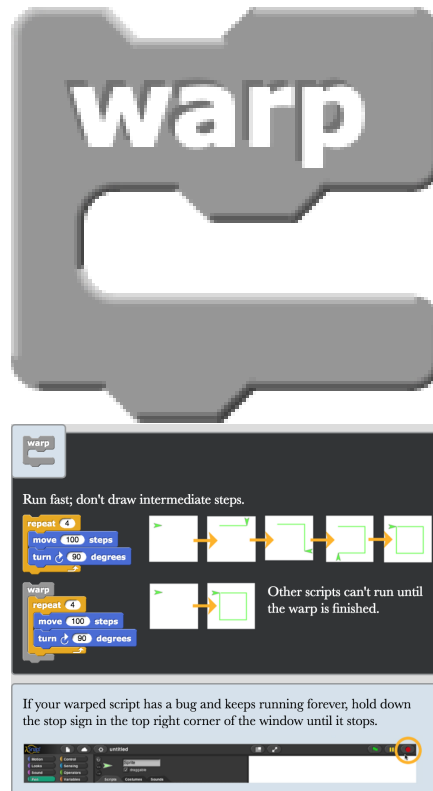
---

No examples yet.

## 0.95 Warp

---

Complete Me



### 0.95.1 Example Images

---

No examples yet.

### 0.95.2 Example Projects

---

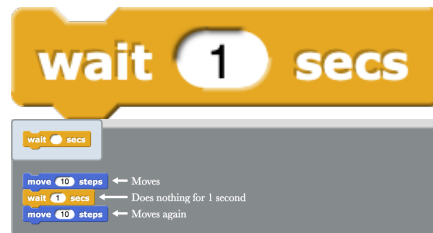
No examples yet.



## 0.96 wait secs

---

Complete Me



### 0.96.1 Example Images

---

No examples yet.

### 0.96.2 Example Projects

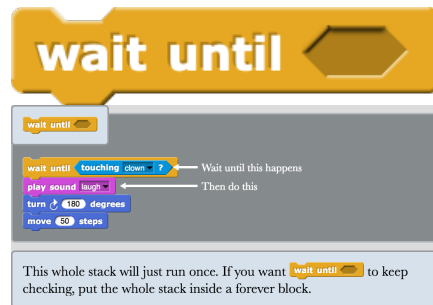
---

No examples yet.

## 0.97 wait until

---

Complete Me



### 0.97.1 Example Images

---

No examples yet.

### 0.97.2 Example Projects

---

No examples yet.

## 0.98 forever

---

Runs the scripts in a loop until the stop button is clicked.



### 0.98.1 Example Images

---

No examples yet.

### 0.98.2 Example Projects

---

No examples yet.

## 0.99 repeat

---

Complete Me



### 0.99.1 Example Images

---

No examples yet.

### 0.99.2 Example Projects

---

No examples yet.

## 0.100 repeat until

---

Complete Me



### 0.100.1 Example Images

---

No examples yet.

### 0.100.2 Example Projects

---

No examples yet.

## 0.101 for \_ = to

---

The for block lets you make a loop with a new variable (i) that starts a 1 and increments by 1 value, up to 10.



### 0.101.1 Example Images

---

No examples yet.

### 0.101.2 Example Projects

---

No examples yet.

## 0.102 if

---

Complete Me



### 0.102.1 Example Images

---

No examples yet.

### 0.102.2 Example Projects

---

No examples yet.

## 0.103 if else

---

Complete Me



### 0.103.1 Example Images

---

No examples yet.

### 0.103.2 Example Projects

---

No examples yet.



## 0.104 if then else

---

Complete Me



### 0.104.1 Example Images

---

No examples yet.

### 0.104.2 Example Projects

---

No examples yet.

## 0.105 report

---

Complete Me



### 0.105.1 Example Images

---

No examples yet.

### 0.105.2 Example Projects

---

No examples yet.

## 0.106 stop

---

Complete Me



### 0.106.1 Example Images

---

No examples yet.

### 0.106.2 Example Projects

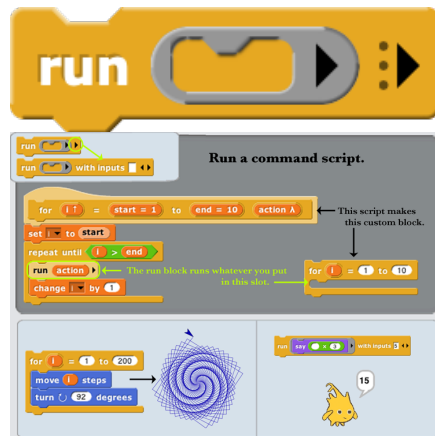
---

No examples yet.

## 0.107 run

---

Complete Me



### 0.107.1 Example Images

---

No examples yet.

### 0.107.2 Example Projects

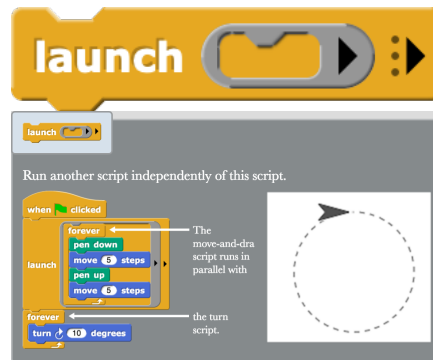
---

No examples yet.

## 0.108 launch

---

Complete Me



### 0.108.1 Example Images

---

No examples yet.

### 0.108.2 Example Projects

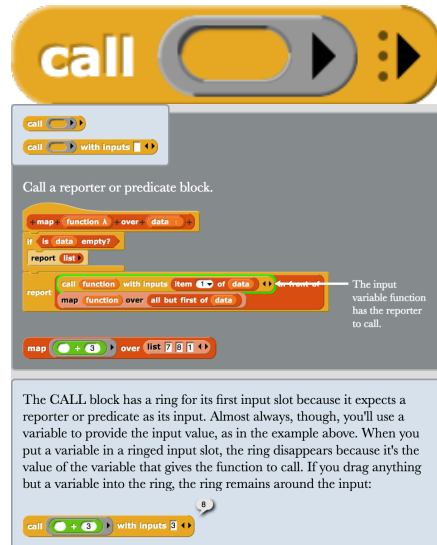
---

No examples yet.

## 0.109 call

---

Complete Me



### 0.109.1 Example Images

---

No examples yet.

### 0.109.2 Example Projects

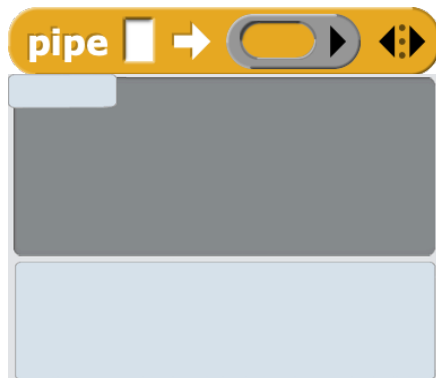
---

No examples yet.

## 0.110 pipe →

---

Complete Me



### 0.110.1 Example Images

---

No examples yet.

### 0.110.2 Example Projects

---

No examples yet.

0.111 tell to

---

Complete Me

tell

to

tell

to

Sprite2

to

show

for each

sprite

in

my other sprites

tell

to

go to center

← Tell a particular other sprite to show itself.

← Tell all sprites except myself to move to the center of the stage. (Each sprite is used, one at a time, to fill the empty slot highlighted in green.)

The script is run in the environment of the other object, so it uses that object's local variables and methods. If you want to use one of the sender's variables inside the script, leave an empty input slot in the script, click the right arrowhead at the end of the TELL block, and use the sender's variable as input to TELL.

repeat

this variable

times

move

steps

turn

degrees

with inputs

my variable

In this example, "this variable" is defined in Sprite2, so it can be used inside the script. "My variable" is defined in the sending sprite, so it can't be used inside the script. Instead it is used as input to TELL, which fills the empty slot in the MOVE block with it.

0.111.1 Example Images

---

No examples yet.

0.111.2 Example Projects

---

No examples yet.

112



# 0.112 ask for

Complete Me

askfor

ask for

ask another object a question.  
Ask a particular sprite where it is:

ask Sprite(3) for list x position y position

|   |     |
|---|-----|
| 1 | 156 |
| 2 | 20  |

length: 2

Ask all sprites except myself where they are:

mapask forlet x position y positionover  
my other sprite

|   |      |     |
|---|------|-----|
| 2 | A    | B   |
| 1 | -177 | -76 |
| 2 | 156  | 20  |

(Each sprite is used, one at a time, to fill the empty slot circled in green.)

The expression is evaluated in the environment of the other object, so it uses that object's local variables and methods. If you want to use one of the sender's variables inside the script, leave an empty input slot in the script, click the right arrowhead at the end of the ASK, and use the sender's variable as input to ASK.

ask Sprite(3) for list its variable with inputs my variable

In this example, "its variable" is defined in Sprite(3), so it can be used inside the expression. "My variable" is defined in the asking sprite, so it can't be used inside the expression. Instead it is used as input to ASK, which fills the empty slot in the LIST block with it.

## 0.112.1 Example Images

No examples yet.

## 0.112.2 Example Projects

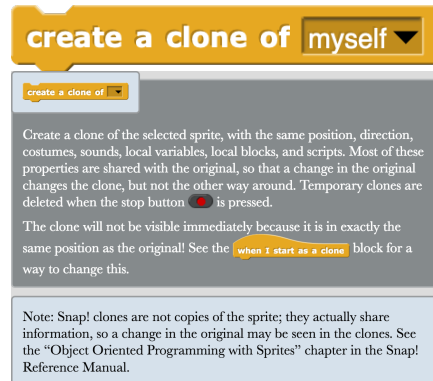
No examples yet.

113

## 0.113 create a clone of

---

Complete Me



### 0.113.1 Example Images

---

No examples yet.

### 0.113.2 Example Projects

---

No examples yet.

## 0.114 a new clone of

---

Complete Me

**a new clone of** myself ▼

**a new clone of** myself ▼

Report a new clone of the selected sprite, with the same position, direction, costumes, sounds, local variables, local blocks, and scripts. Most of these properties are shared with the original, so that a change in the original changes the clone, but not the other way around. Temporary clones are deleted when the stop button  is pressed.

The clone will not be visible immediately because it is in exactly the same position as the original! See the **when I start as a clone** block for a way to change this.

Note: Snap! clones are not copies of the sprite; they actually share information, so a change in the original may be seen in the clones. See the “Object Oriented Programming with Sprites” chapter in the Snap! Reference Manual.

### 0.114.1 Example Images

---

No examples yet.

### 0.114.2 Example Projects

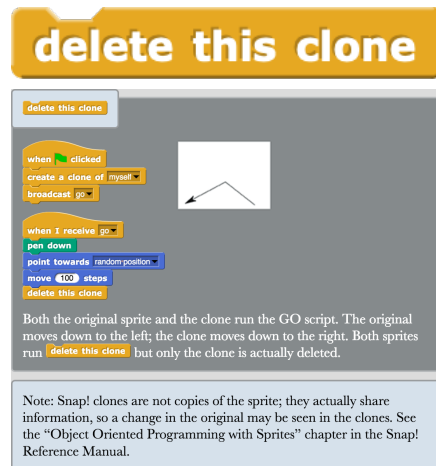
---

No examples yet.

## 0.115 delete this clone

---

Complete Me



### 0.115.1 Example Images

---

No examples yet.

### 0.115.2 Example Projects

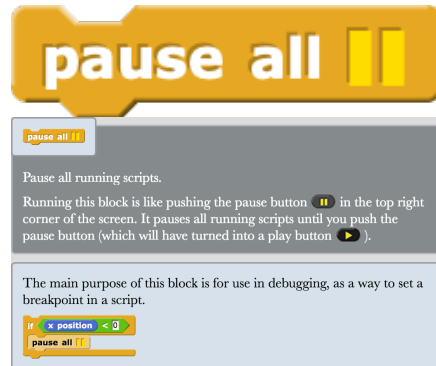
---

No examples yet.

## 0.116 pause all

---

Complete Me



### 0.116.1 Example Images

---

No examples yet.

### 0.116.2 Example Projects

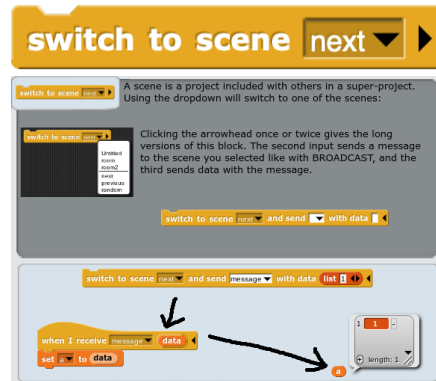
---

No examples yet.

## 0.117 switch to scene

---

Complete Me



### 0.117.1 Example Images

---

No examples yet.

### 0.117.2 Example Projects

---

No examples yet.

## 0.118 define

---

Complete Me



### 0.118.1 Example Images

---

No examples yet.

### 0.118.2 Example Projects

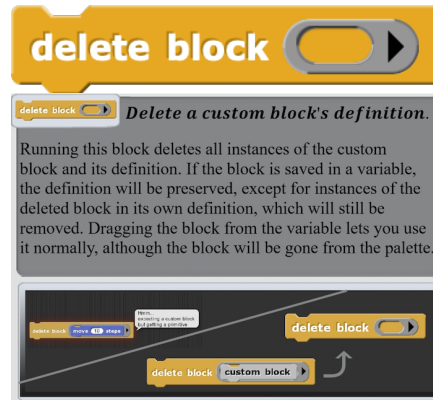
---

No examples yet.

## 0.119 delete block

---

Complete Me



### 0.119.1 Example Images

---

No examples yet.

### 0.119.2 Example Projects

---

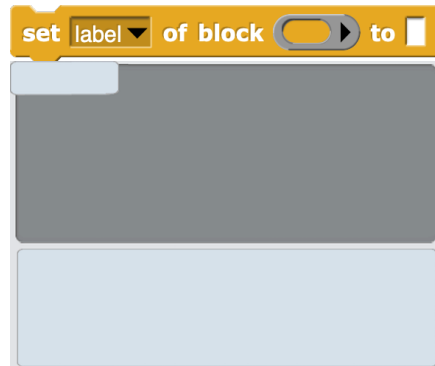
No examples yet.



## 0.120 set of block to

---

Complete Me



### 0.120.1 Example Images

---

No examples yet.

### 0.120.2 Example Projects

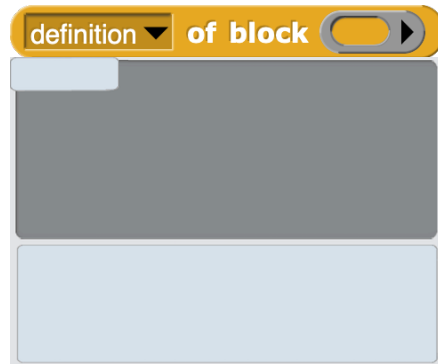
---

No examples yet.

## 0.121 Attribute Of Block

---

Complete Me



### 0.121.1 Example Images

---

No examples yet.

### 0.121.2 Example Projects

---

No examples yet.

## 0.122 this

---

Complete Me



### 0.122.1 Example Images

---

No examples yet.

### 0.122.2 Example Projects

---

No examples yet.

## 0.123 set slot to

---

Complete Me



### 0.123.1 Example Images

---

No examples yet.

### 0.123.2 Example Projects

---

No examples yet.

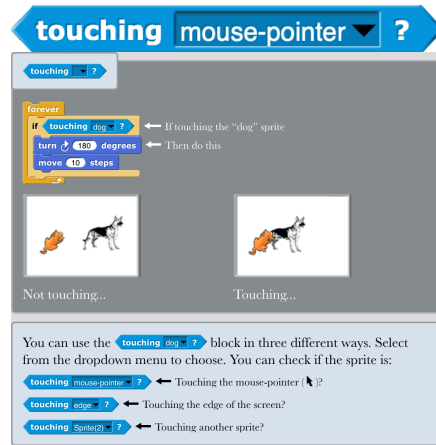
## 0.124 Sensing Blocks

---

## 0.125 touching ?

---

Complete Me



### 0.125.1 Example Images

---

No examples yet.

### 0.125.2 Example Projects

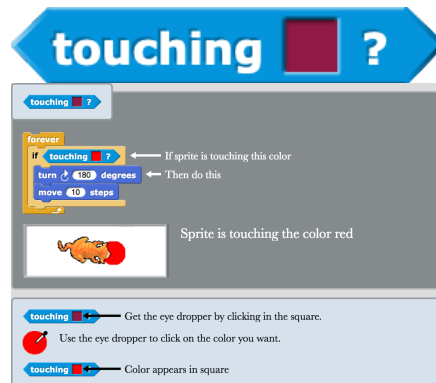
---

No examples yet.

## 0.126 touching ?

---

Complete Me



### 0.126.1 Example Images

---

No examples yet.

### 0.126.2 Example Projects

---

No examples yet.

0.127 color is touching ?

---

Complete Me

color is touching ?

color is touching ?

forever

if color is touching ?

play note 40 for 0.5 beats

not touching

touching

color is touching

Get the eye dropper by clicking in the square.

Use the eye dropper to click on the color you want.

color is touching

Color appears in square

0.127.1 Example Images

---

No examples yet.

0.127.2 Example Projects

---

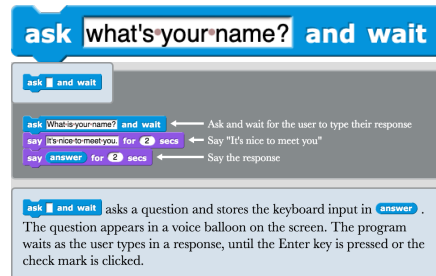
No examples yet.



## 0.128 ask and wait

---

Complete Me



### 0.128.1 Example Images

---

No examples yet.

### 0.128.2 Example Projects

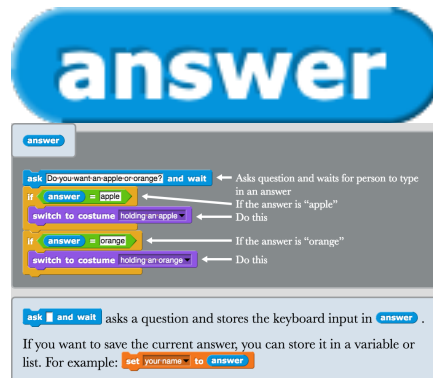
---

No examples yet.

## 0.129 answer

---

Complete Me



### 0.129.1 Example Images

---

No examples yet.

### 0.129.2 Example Projects

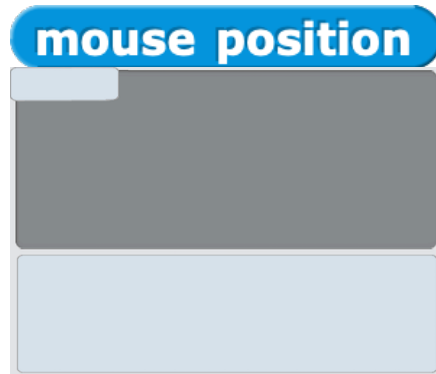
---

No examples yet.

## 0.130 mouse position

---

Complete Me



### 0.130.1 Example Images

---

No examples yet.

### 0.130.2 Example Projects

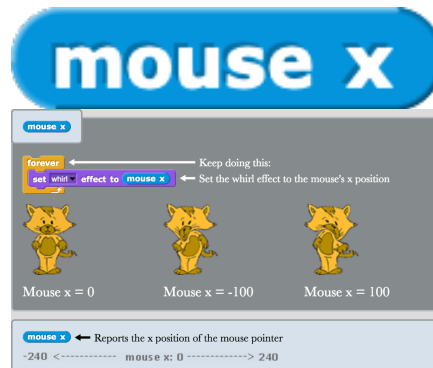
---

No examples yet.

## 0.131 mouse x

---

Complete Me



### 0.131.1 Example Images

---

No examples yet.

### 0.131.2 Example Projects

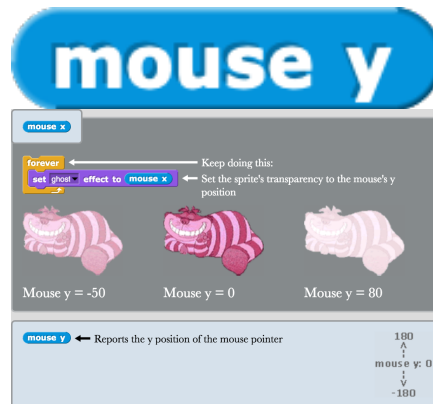
---

No examples yet.

## 0.132 mouse y

---

Complete Me



### 0.132.1 Example Images

---

No examples yet.

### 0.132.2 Example Projects

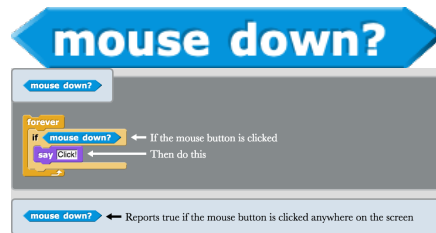
---

No examples yet.

## 0.133 mouse down?

---

Complete Me



### 0.133.1 Example Images

---

No examples yet.

### 0.133.2 Example Projects

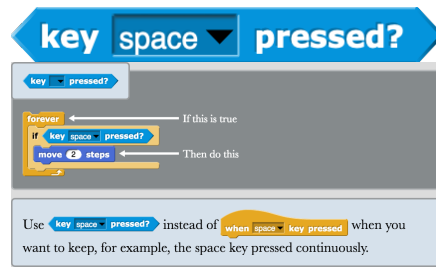
---

No examples yet.

## 0.134 key pressed?

---

Complete Me



### 0.134.1 Example Images

---

No examples yet.

### 0.134.2 Example Projects

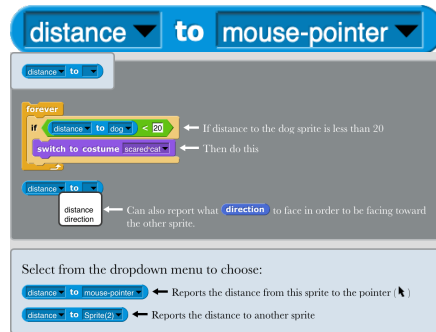
---

No examples yet.

## 0.135 Distance To

---

Complete Me



### 0.135.1 Example Images

---

No examples yet.

### 0.135.2 Example Projects

---

No examples yet.





## 0.137 reset timer

---

Complete Me



### 0.137.1 Example Images

---

No examples yet.

### 0.137.2 Example Projects

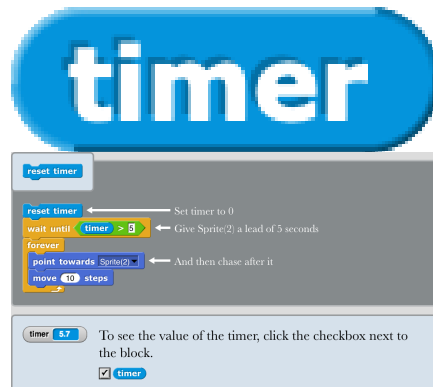
---

No examples yet.

## 0.138 timer

---

Complete Me



### 0.138.1 Example Images

---

No examples yet.

### 0.138.2 Example Projects

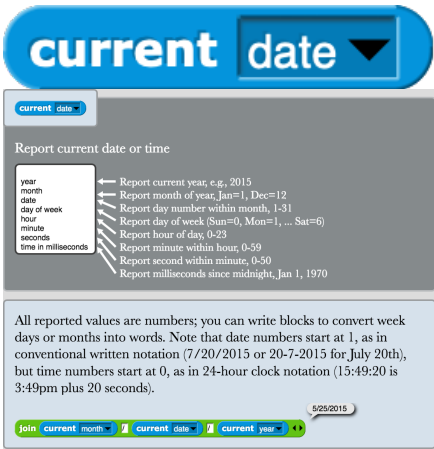
---

No examples yet.

0.139 current

---

Complete Me



0.139.1 Example Images

---

No examples yet.

0.139.2 Example Projects

---

No examples yet.

## 0.140 Attribute Of

---

Complete Me



### 0.140.1 Example Images

---

No examples yet.

### 0.140.2 Example Projects

---

No examples yet.

## 0.141 my

---

Complete Me



### 0.141.1 Example Images

---

No examples yet.

### 0.141.2 Example Projects

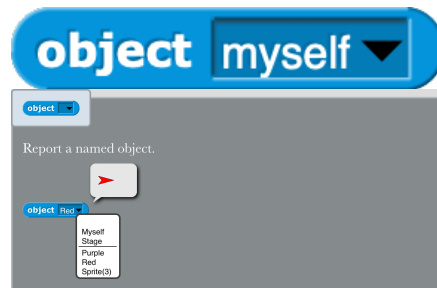
---

No examples yet.

## 0.142 object

---

Complete Me



### 0.142.1 Example Images

---

No examples yet.

### 0.142.2 Example Projects

---

No examples yet.

## 0.143 url

---

Complete Me

**url** snap.berkeley.edu

Report web page text.

```
<html>
<head>
<meta name="snap-cloud-domain" location="https://snap.berkeley.edu:443">
<meta charset="UTF-8">
<title>Snap! Build Your Own Blocks</title>
<meta name="description" content="The Snap! social platform">
<meta name="author" content="Bernat Romagosa">
<meta name="snap-cloud-domain" location="https://snap.berkeley.edu/">
<link rel="icon" href="favicon.ico">
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Hint: Use SPLIT to see the result as a list of lines.

split url snap.berkeley.edu by line

Note: Many websites won't work with this block because of an Internet security feature built into browsers, called "Cross Origin Resource Sharing." There's nothing Snap! can do about this; you should try to get the provider of the website to allow cross-origin requests from everyone, or at least for your Internet domain. You can also look for a "CORS proxy," which is a website that funnels requests for arbitrary pages regardless of CORS status.

### 0.143.1 Example Images

---

No examples yet.

### 0.143.2 Example Projects

---

No examples yet.



# 0.144 microphone

Complete Me

microphone volume

microphone volume

volume

note

frequency

samples

spectrum

resolution

The first three options report a single number characterizing the sound as a whole average volume, dominant note number, and frequency.

The next two options are time domain information: a list of instantaneous volumes and a number representing the number of samples per second (sampling rate).

The last two options are frequency domain information: a histogram of frequencies and the number of buckets in the histogram.

Report data from microphone.

This block collects and analyzes a short (0.01 to 0.1 second) burst of sound.

“Time domain” means that the list items are the instantaneous volume at each moment, measured thousands of times per second. “Frequency domain” means that the list items are the volume over the entire burst of sound at each note frequency.

time domain



freq. domain



time domain



freq. domain



Tuning fork (pure sine wave)

Simple musical instrument with harmonics

## 0.144.1 Example Images

No examples yet.

## 0.144.2 Example Projects

No examples yet.

0.145 video on

---

Complete Me

video

motion

on

myself

video

motion

on

myself

Reports information from video capture.

The second input is a menu of sprites, and restricts the block's attention to the part of the stage taken up by the chosen sprite. ("Stage" uses the entire stage area.)

The first input specifies what information you want:

video

motion

on

myself

snapshot

motion

direction

Reports a snapshot of the video as a costume.

Reports a number, the speed of a detected moving object in the video.

Reports a number, the direction in which the object is moving, suitable for use as input in POINT IN DIRECTION.

0.145.1 Example Images

---

No examples yet.

0.145.2 Example Projects

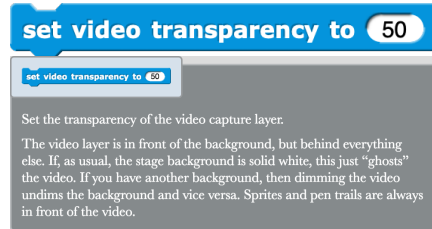
---

No examples yet.

## 0.146 set video transparency to

---

Complete Me



### 0.146.1 Example Images

---

No examples yet.

### 0.146.2 Example Projects

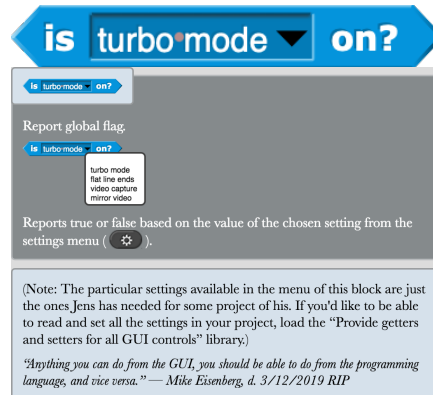
---

No examples yet.

## 0.147 is on?

---

Complete Me



### 0.147.1 Example Images

---

No examples yet.

### 0.147.2 Example Projects

---

No examples yet.

## 0.148 set to

---

Complete Me



### 0.148.1 Example Images

---

No examples yet.

### 0.148.2 Example Projects

---

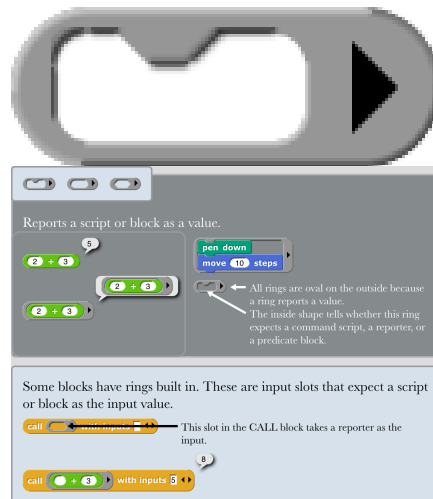
No examples yet.



## 0.150 Command Ring

---

Complete Me



### 0.150.1 Example Images

---

No examples yet.

### 0.150.2 Example Projects

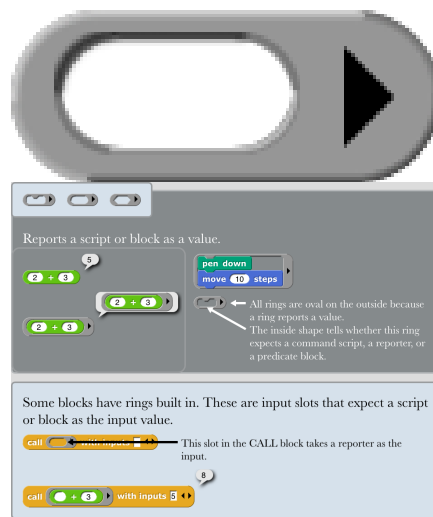
---

No examples yet.

## 0.151 Reporter Ring

---

Complete Me



### 0.151.1 Example Images

---

No examples yet.

### 0.151.2 Example Projects

---

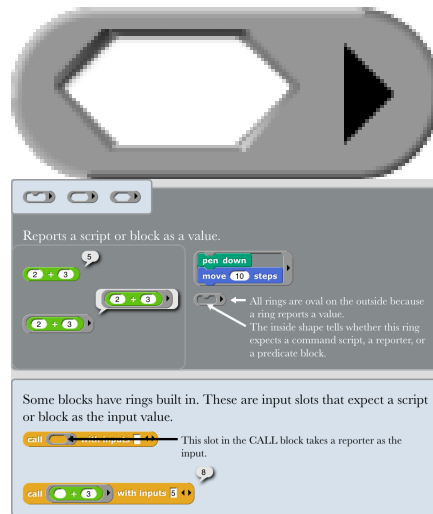
No examples yet.



## 0.152 Predicate Ring

---

Complete Me



### 0.152.1 Example Images

---

No examples yet.

### 0.152.2 Example Projects

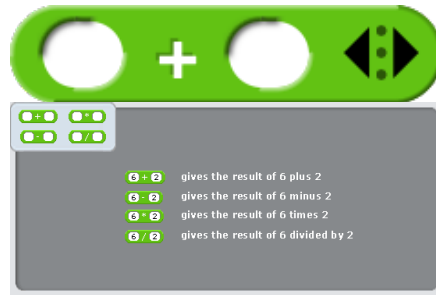
---

No examples yet.

## 0.153 Sum +

---

Complete Me



### 0.153.1 Example Images

---

No examples yet.

### 0.153.2 Example Projects

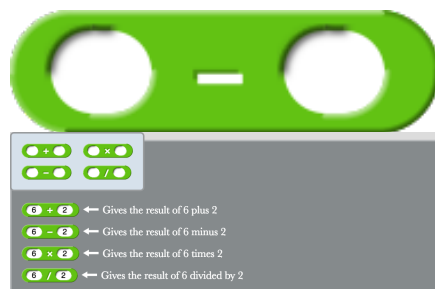
---

No examples yet.

## 0.154 Subtract -

---

Complete Me



### 0.154.1 Example Images

---

No examples yet.

### 0.154.2 Example Projects

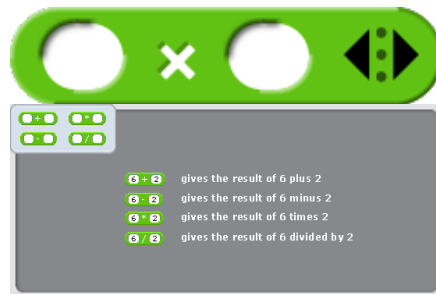
---

No examples yet.

## 0.155 Product ×

---

Complete Me



### 0.155.1 Example Images

---

No examples yet.

### 0.155.2 Example Projects

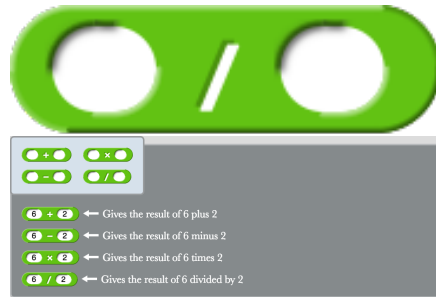
---

No examples yet.

## 0.156 Divide ÷

---

Complete Me



### 0.156.1 Example Images

---

No examples yet.

### 0.156.2 Example Projects

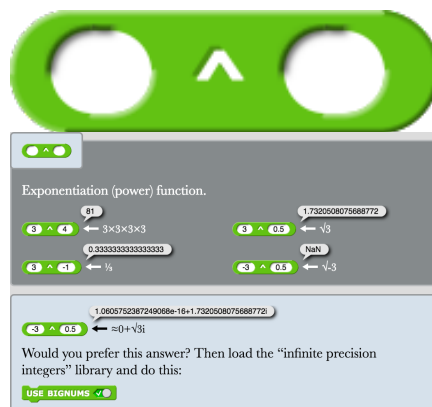
---

No examples yet.

## 0.157 Power of Number

---

Complete Me



### 0.157.1 Example Images

---

No examples yet.

### 0.157.2 Example Projects

---

No examples yet.

0.158 Mod

---

Complete Me

mod

mod

forever

move 1 steps

if x position mod 10 = 0

change y by 10

wait 0.3 sec

change y by 10

Keep doing this

Take one step

Every ten steps

Jump

mod gives the remainder of the division of the two numbers.

For example, 143 mod 6 is 5 (143 divided 6 is 23 and the remainder is 5).

0.158.1 Example Images

---

No examples yet.

0.158.2 Example Projects

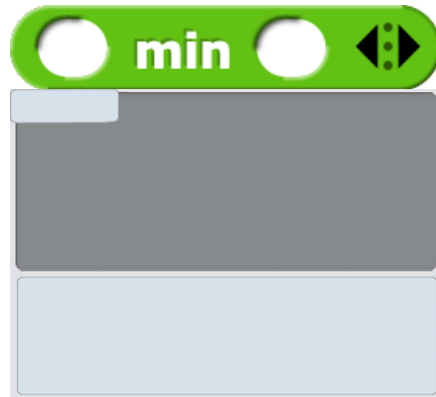
---

No examples yet.

## 0.159 min

---

Complete Me



### 0.159.1 Example Images

---

No examples yet.

### 0.159.2 Example Projects

---

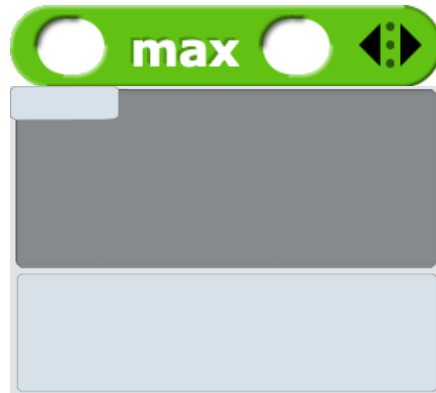
No examples yet.



## 0.160 max

---

Complete Me



### 0.160.1 Example Images

---

No examples yet.

### 0.160.2 Example Projects

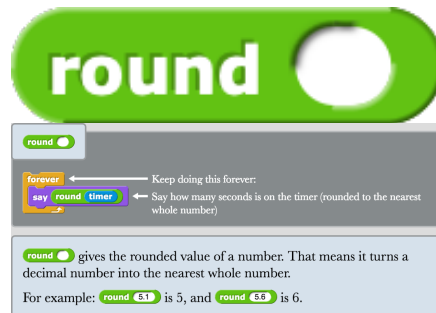
---

No examples yet.

## 0.161 round

---

Complete Me



### 0.161.1 Example Images

---

No examples yet.

### 0.161.2 Example Projects

---

No examples yet.

0.162 Math Functions

---

Complete Me

sqrt of 10

sqrt of

pen down  
forever  
change x by  
set y to

Put the pen down  
Forever  
Move to the right  
Set the y position to a function of the x position

Set y based on the sine of x

sqrt of

abs  
neg  
ceil  
floor  
sqrt  
sin  
cos  
tan  
asin  
acos  
atan

This block performs a mathematical function. Choose the math function from the block's drop down menu.  
Note: Trig functions are based on angles in degrees, not radians!

0.162.1 Example Images

---

No examples yet.

0.162.2 Example Projects

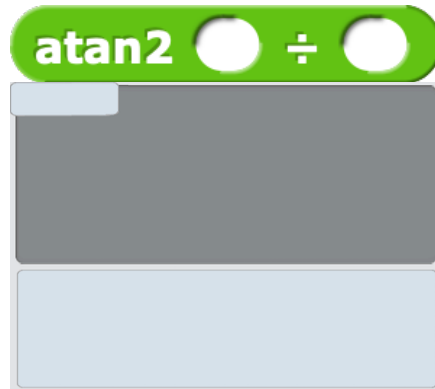
---

No examples yet.

## 0.163 atan2 ÷

---

Complete Me



### 0.163.1 Example Images

---

No examples yet.

### 0.163.2 Example Projects

---

No examples yet.

## 0.164 pick random to

---

Complete Me



### 0.164.1 Example Images

---

No examples yet.

### 0.164.2 Example Projects

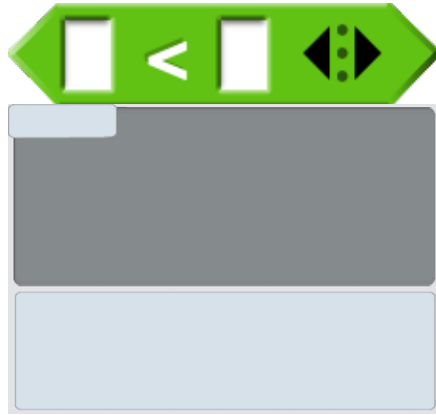
---

No examples yet.

## 0.165 <

---

Complete Me



### 0.165.1 Example Images

---

No examples yet.

### 0.165.2 Example Projects

---

No examples yet.

## 0.166 \_

---

Complete Me



### 0.166.1 Example Images

---

No examples yet.

### 0.166.2 Example Projects

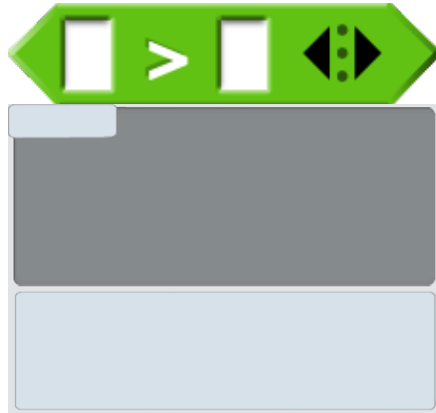
---

No examples yet.

## 0.167 >

---

Complete Me



### 0.167.1 Example Images

---

No examples yet.

### 0.167.2 Example Projects

---

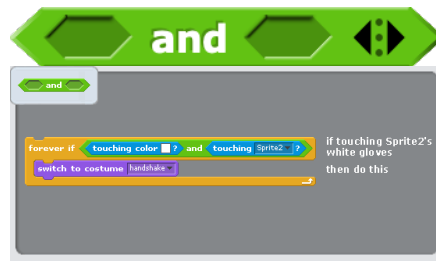
No examples yet.



## 0.168 \_

---

Complete Me



### 0.168.1 Example Images

---

No examples yet.

### 0.168.2 Example Projects

---

No examples yet.

## 0.169 or

---

Complete Me



### 0.169.1 Example Images

---

No examples yet.

### 0.169.2 Example Projects

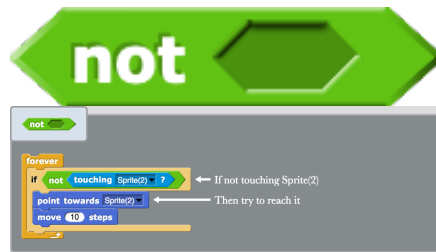
---

No examples yet.

## 0.170 not

---

Complete Me



### 0.170.1 Example Images

---

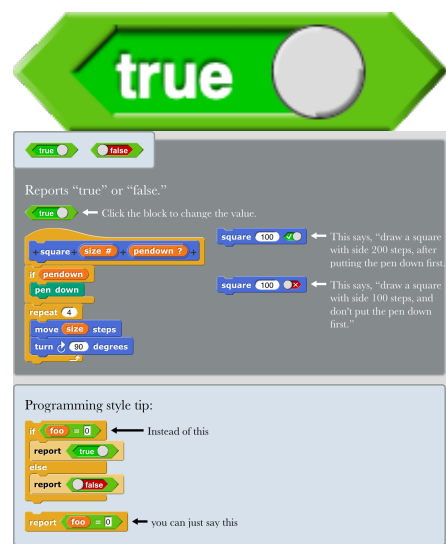
No examples yet.

### 0.170.2 Example Projects

---

No examples yet.

Complete Me



0.171.1 Example Images

---

No examples yet.

0.171.2 Example Projects

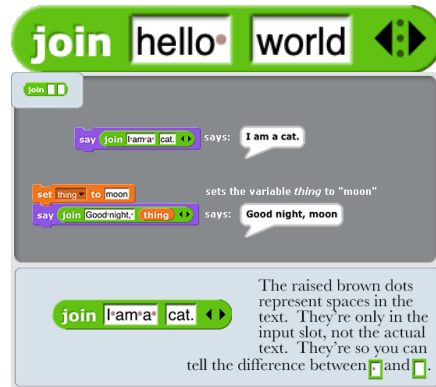
---

No examples yet.

## 0.172 join

---

Complete Me



### 0.172.1 Example Images

---

No examples yet.

### 0.172.2 Example Projects

---

No examples yet.

0.173 split by

Complete Me

split

hello-world

by

split hello-world by

Report list of text split into words or lines.

1 nowhere

2

3

4

5

6

7

length: 4

split hello-world by

Splitting on a character (space, in this example) can result in empty list elements if the character appears several times in a row in the text.

1 nowhere

2

3

4

5

6

7

length: 2

split nowhere-man by whitespace

Choosing "whitespace" from the pulldown menu will treat any number of spaces, tabs, or newlines as a single separator, to make a list of words.

split yahoo by letter

Split a text into individual characters (including spaces, newlines, and punctuation as list items).

332 items

1 <html>

2 <head>

3 <meta name="snap-o

4 <meta charset="U

5 <title>Snap! Ban

6 <meta name="des

7 <meta name="s

split url snap.berkeley.edu by line

Split a multiline text into lines. (The newline character, the carriage return character, or the combination CR, NL all count as a single line break in the text.)

1 b

2

3

4

5

6

7

length: 3

split banana by an

It's not often useful, but you can use a multi-character string as the separator. In this example, we get an empty list item because "an" appears twice in a row in "banana."

0.173.1 Example Images

No examples yet.

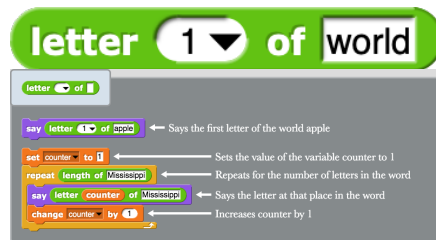
0.173.2 Example Projects

No examples yet.

## 0.174 letter of

---

Complete Me



### 0.174.1 Example Images

---

No examples yet.

### 0.174.2 Example Projects

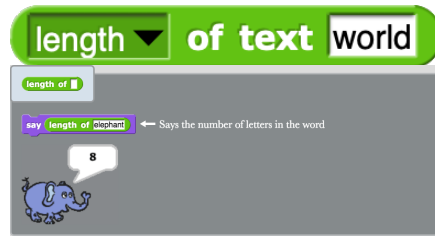
---

No examples yet.

## 0.175 Attribute of Text

---

Complete Me



### 0.175.1 Example Images

---

No examples yet.

### 0.175.2 Example Projects

---

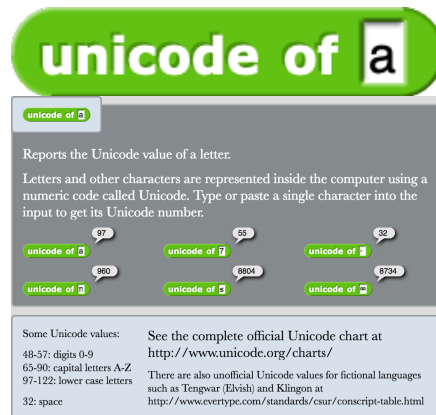
No examples yet.



## 0.176 unicode of

---

Complete Me



### 0.176.1 Example Images

---

No examples yet.

### 0.176.2 Example Projects

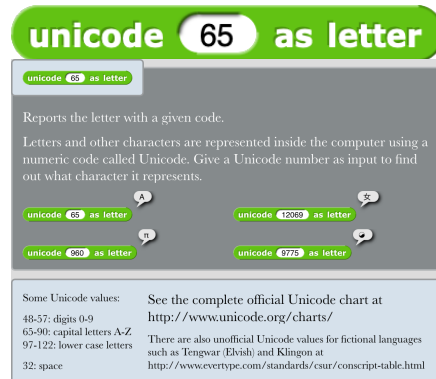
---

No examples yet.

## 0.177 unicode as letter

---

Complete Me



### 0.177.1 Example Images

---

No examples yet.

### 0.177.2 Example Projects

---

No examples yet.

0.178 is a ?

---

Complete Me

is 5 a number ?

is 5 a number ?

Reports true if input is a given type.

All of these are true:

is 537 a number ?

is a text ?

is 3 + 4 a number ?

is true a text ?

is 3 + 4 a reporter ?

is true a Boolean ?

is pen up a command ?

is true a predicate ?

The possible types are number, text, Boolean (true or false), list, sprite, costume, sound (object-oriented types), command (a script or jigsaw-shaped command block), reporter (an oval block that reports a value), and predicate (a hexagonal block that reports true or false).  
Every value has exactly one of these types.

is a number ?

number  
text  
Boolean  
list  
sprite  
costume  
sound  
command  
reporter  
predicate

0.178.1 Example Images

---

No examples yet.

0.178.2 Example Projects

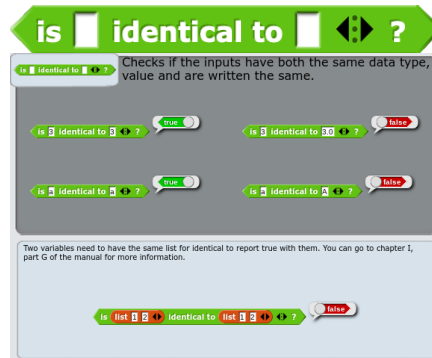
---

No examples yet.

## 0.179 is ?

---

Complete Me



### 0.179.1 Example Images

---

No examples yet.

### 0.179.2 Example Projects

---

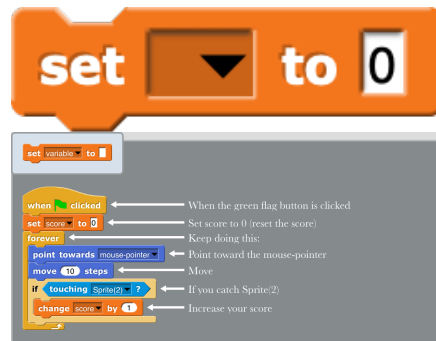
No examples yet.



## 0.181 set to

---

Complete Me



### 0.181.1 Example Images

---

No examples yet.

### 0.181.2 Example Projects

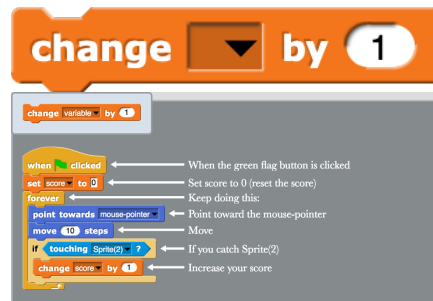
---

No examples yet.

## 0.182 change by

---

Complete Me



### 0.182.1 Example Images

---

No examples yet.

### 0.182.2 Example Projects

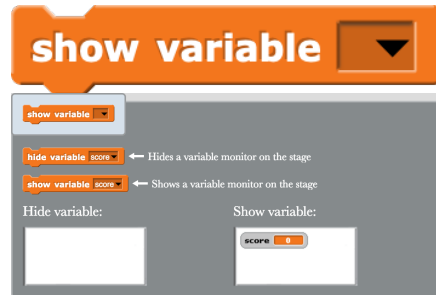
---

No examples yet.

## 0.183 show variable

---

Complete Me



### 0.183.1 Example Images

---

No examples yet.

### 0.183.2 Example Projects

---

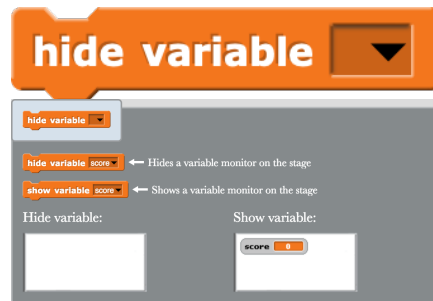
No examples yet.



## 0.184 hide variable

---

Complete Me



### 0.184.1 Example Images

---

No examples yet.

### 0.184.2 Example Projects

---

No examples yet.

## 0.185 Script variables

---

Complete Me

script variables **a**

script variables **a**

swap items index1 # and index2 # in list

script variables temp

set temp to item index1 of list

replace item index1 of list with item index2 of list

replace item index2 of list with temp

Every time this block is called, it makes a new temporary variable **temp**. Click (don't drag) to change name.

Script variables don't clutter up your palette, and you can be sure no other block changes their value. This works for scripting arca scripts, too, not just custom blocks.

new counter

script variables count

set count to 0

report change count by 1

report report count

Object Oriented Programming:  
If a reporter reports a script, and that script uses a script variable created outside the reported script (such as the use of **count** in the script that **new counter** reports) then the script variable isn't temporary!

set mycounter to new counter

say call my counter for 2 secs → 1

say call my counter for 2 secs → 2

### 0.185.1 Example Images

---

No examples yet.

### 0.185.2 Example Projects

---

No examples yet.

0.186 inherit

---

Complete Me

inherit

inherit

Share a property with your parent. (This only works for clones.)

BEFORE

Run

AFTER

x position

y position

direction

inherit x position

x position

y position

direction

Parent

Clone

Clone's x position is now tied to parent's.

Pale background in palette for inherited property.

Inheritable properties include these:

sprite-local variable

x position

y position

direction

size

costumes

costume #

sounds

scripts

sprite-local variable

(and more... See Reference Manual Chapter VII.)

You can also control inheritance through the user user interface. Right-click on a shareable thing in the palette and see this:

costume #

help...

hide

inherited

Click to inherit

0.186.1 Example Images

---

No examples yet.

0.186.2 Example Projects

---

No examples yet.

## 0.187 list

---

Complete Me



### 0.187.1 Example Images

---

No examples yet.

### 0.187.2 Example Projects

---

No examples yet.

## Complete Me



No examples yet.

No examples yet.

0.189 In Front Of

Complete Me

in front of

in front of

Report a list that extends another list with one new item.

in front of

The list you want to extend  
New item (can be anything!)

2 in front of list 3 1

1 2 3  
2 8 2  
3 1 2  
length: 3

Note: The new list shares structure with the old one!

set foo to list 3 1

set baz to 2 in front of foo

is foo identical to all but first of baz

true

initials data

if is data empty?

report letter 1 of item 1 of data in front of initials all but first of data

some function item 1 of foo in front of recursive call all but first of foo

1 T  
2 L  
3 A  
length: 3

initials list three letter Acronym

in front of

is most useful in writing recursive reporters like this one, to build up a list element by element.

0.189.1 Example Images

No examples yet.

0.189.2 Example Projects

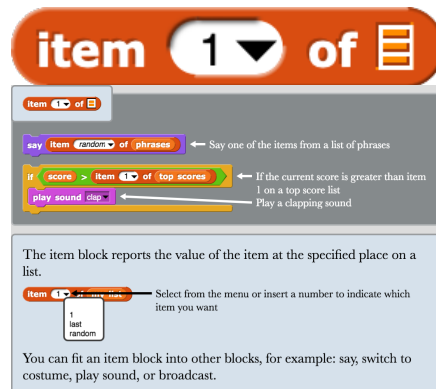
No examples yet.

190

## 0.190 item of

---

Complete Me



### 0.190.1 Example Images

---

No examples yet.

### 0.190.2 Example Projects

---

No examples yet.

0.191 all but first of

---

Complete Me

all but first of

all but first of

Report all but first list item.

18

21

length: 2

all but first of list 7 3 1

The new list shares structure with the old one; it's not a copy. If the input list was made with the list reporter blocks **In front of** **Item** **of** **all but first of** then **all but first of** is very fast; it doesn't have to copy the items or the list structure. If you use list commands such as **Insert item at** **of** on a list, then it will be a a dynamic array rather than a linked list, so this block has to convert to linked form.

The list reporters are good for writing recursive list reporters that build up a result list item by item.

0.191.1 Example Images

---

No examples yet.

0.191.2 Example Projects

---

No examples yet.



# 0.192 Report List Attribute

Complete Me

length of

length of

length of list 3 3 3 4

Reports an attribute of the list

length of list 3 3 3 4

It gives the number of items in the *automat* list.

length of list 3 list 3 3 3 4

Here are the possible attributes:

length  
rank  
arrays one  
flatten  
columns  
reverse

LENGTH is described above  
RANK reports the LENGTH OF DIMENSIONS OF its input  
DIMENSIONS reports how long the list is in each dimension  
FLATTEN reports each non-list item, even those from sublists.  
E.g., FLATTEN OF (a (b c) d e f (g h)) gives (a b c d e f g)  
COLLIMNS, given (a b c) (d e f), reports (a d) (b e) (c f)  
REVERSE, given (a b (c d e) i), reports f (e d c) b a)

These three report texts, not lists:  
LINPS joins each item with a newline  
CSV takes a 2D list and gives the spreadsheet equivalent  
JSON takes an arbitrarily deep list, often with key-value pairs, and gives the JSON equivalent

array  
list  
not

## 0.192.1 Example Images

No examples yet.

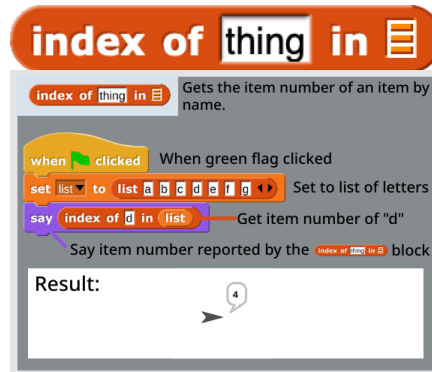
## 0.192.2 Example Projects

No examples yet.

## 0.193 index of in

---

Complete Me



### 0.193.1 Example Images

---

No examples yet.

### 0.193.2 Example Projects

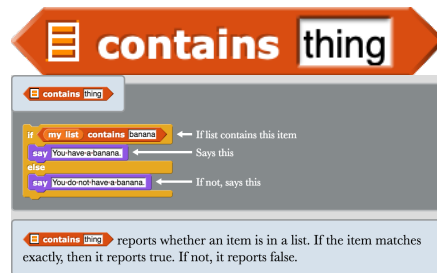
---

No examples yet.

## 0.194 List Contains

---

Complete Me



### 0.194.1 Example Images

---

No examples yet.

### 0.194.2 Example Projects

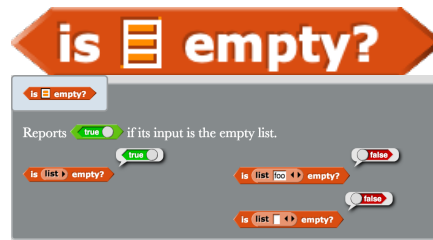
---

No examples yet.

## 0.195 is empty?

---

Complete Me



### 0.195.1 Example Images

---

No examples yet.

### 0.195.2 Example Projects

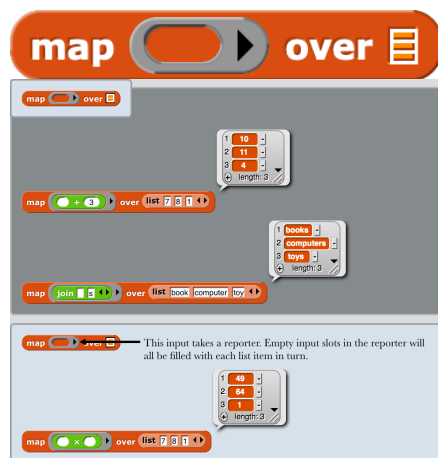
---

No examples yet.

## 0.196 map over

---

Complete Me



### 0.196.1 Example Images

---

No examples yet.

### 0.196.2 Example Projects

---

No examples yet.

## 0.197 keep items from

---

Complete Me



The image shows the 'keep items from' block in a Scratch-like environment. The block has a title bar with 'keep items' and 'from' and a list icon. Below the title bar, there are two examples of how to use the block. The first example shows the 'keep items' block with a 'is a number' predicate and a 'from' block containing a list of items: 'cat', 'dog', 'fish', 'hen', 'horse', 'turtle', 'dove'. The second example shows the 'keep items' block with a 'length of' predicate and a 'from' block containing a list of items: 'University of California at Berkeley'. Below the examples, there is a text box explaining the block's behavior.

Each list item is put into the empty input slot of the predicate block. If the predicate reports **true** then that item is included in the result list. Note that if the predicate has a default input value, you have to clear the slot to use it with KEEP; e.g., this won't work:

KEEP makes a new list; the original list is not modified.

### 0.197.1 Example Images

---

No examples yet.

### 0.197.2 Example Projects

---

No examples yet.

## 0.198 find first item in

---

Complete Me

**find first item**  **in** 

**find first item**  **in** 

Reports the first item in the list that matches the predicate.

**find first item**  **length of**  **>**  **in** 

**list**                             

Compare:

|   |            |
|---|------------|
| 1 | University |
| 2 | California |
| 3 | Berkeley   |
| 4 | length: 3  |

**keep items**  **length of**  **>**  **from** 

**list**                             

**find first item**  **=**  **in** 

**list**                             

**keep items**  **=**  **from** 

**list**                             

FIND FIRST reports False if no item in the list matches the predicate. (KEEP reports the empty list in that situation.)

FIND FIRST means the same as **item**  **of** **keep items**  **from** 

but FIND FIRST is faster if a match is found because it doesn't search the rest of the list.

### 0.198.1 Example Images

---

No examples yet.

### 0.198.2 Example Projects

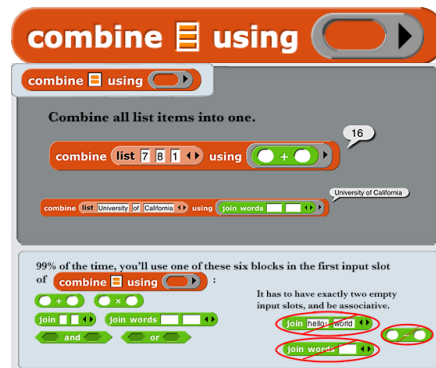
---

No examples yet.

## 0.199 combine using

---

Complete Me



### 0.199.1 Example Images

---

No examples yet.

### 0.199.2 Example Projects

---

No examples yet.



## 0.200 for each in

---

Complete Me



### 0.200.1 Example Images

---

No examples yet.

### 0.200.2 Example Projects

---

No examples yet.

## 0.201 add to

---

Complete Me



### 0.201.1 Example Images

---

No examples yet.

### 0.201.2 Example Projects

---

No examples yet.

# 0.202 delete of

---

Complete Me

delete1 of

delete of

say item of supplies

delete last of supplies

Before:

After:

supplies

water

piece of fruit

biscuits

length: 3

supplies

water

piece of fruit

length: 2

You can specify the number of the item you want to delete. For example:

delete 2 of my list

To delete the last item in the list, choose "last" from the dropdown menu:

delete last of my list

You can also choose to delete everything in the list:

delete all of my list

## 0.202.1 Example Images

---

No examples yet.

## 0.202.2 Example Projects

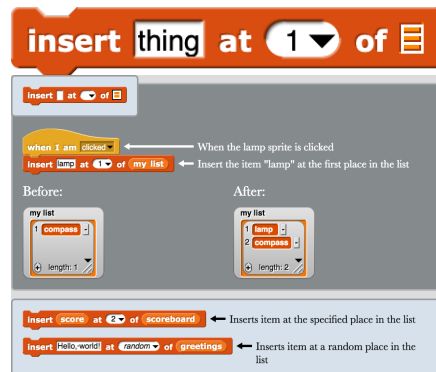
---

No examples yet.

## 0.203 insert at of

---

Complete Me



### 0.203.1 Example Images

---

No examples yet.

### 0.203.2 Example Projects

---

No examples yet.

# 0.204 replace item of with

---

Complete Me



## 0.204.1 Example Images

---

No examples yet.

## 0.204.2 Example Projects

---

No examples yet.

## 0.205 append

---

Complete Me

**append**  

**append**  Joins two lists together.

APPEND reports a new list containing the items from the input lists, in order.

Comparison:

Append



List

|   | A | B | C |
|---|---|---|---|
| 1 | a | b | c |
| 2 | 1 | 2 | 3 |

### 0.205.1 Example Images

---

No examples yet.

### 0.205.2 Example Projects

---

No examples yet.

0.206 reshape to

Complete Me

reshape to 4 3

Reshapes a list to the provided dimensions

Examples:

reshape numbers from 1 to 10 to 5 2

|   |   |    |
|---|---|----|
| 5 | A | B  |
| 1 | 1 | 2  |
| 2 | 3 | 4  |
| 3 | 5 | 6  |
| 4 | 7 | 8  |
| 5 | 9 | 10 |

If the dimensions multiply to a number larger than the length of the list, it starts over with the beginning of the list:

reshape numbers from 1 to 10 to 4 3

|   |   |    |   |   |
|---|---|----|---|---|
| 4 | A | B  | C | D |
| 1 | 1 | 2  | 3 | 4 |
| 2 | 5 | 6  | 7 | 8 |
| 3 | 9 | 10 | 1 | 2 |
| 4 | 3 | 4  | 5 | 6 |

You can use a simple scalar input (number or text string) too:

reshape join 5 to 4 3

|   |   |   |   |
|---|---|---|---|
| 4 | A | B | C |
| 1 | 5 | 5 | 5 |
| 2 | 5 | 5 | 5 |
| 3 | 5 | 5 | 5 |
| 4 | 5 | 5 | 5 |

And with no dimensions, it returns a scalar:

reshape numbers from 1 to 10 to

0.206.1 Example Images

No examples yet.

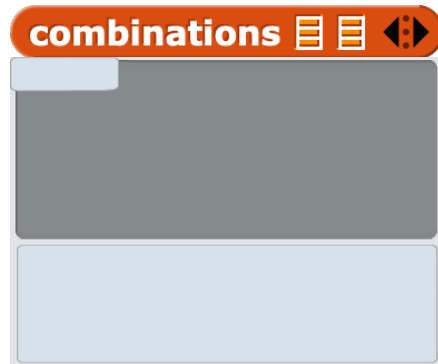
0.206.2 Example Projects

No examples yet.

## 0.207 combinations

---

Complete Me



### 0.207.1 Example Images

---

No examples yet.

### 0.207.2 Example Projects

---

No examples yet.